

INCIDENT MANAGEMENT

EmComm Field Server v1.0

COMPLETE USER MANUAL

McHenry County RACES / ARES / Starcom

Emergency Services Volunteers · Station K9ESV

- Net Control & Starcom Logging
- APRS Tactical & Resource Maps
- Member Roster & Identity System
- NTS, ICS-213, ICS-214 Forms
- Full ICS Platform (5 Sections)
- Offline Maps, Kiwix & FCC Database
- Reference Library & Print Center
- Weather Alerts & HF Propagation

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1 Introduction & System Overview

Incident Management is a self-contained emergency communications server that runs on a Raspberry Pi. It provides a complete suite of EmComm tools — net control logging, an ICS incident management platform, mapping, reference libraries, and form generation — all accessible through a standard web browser with no internet connection required. It is built for McHenry County Emergency Services Volunteers and McHenry County Emergency Management Agency operating under the RACES / ARES / Starcom framework.

How It Works

The Pi broadcasts its own Wi-Fi network named **EMCOMM-NET**. Any smartphone, tablet, or laptop connects to that network and opens the dashboard in a browser at **http://192.168.50.1**. The server hosts every tool locally. No app needs to be installed on any device, and no internet is needed for core functions.

What Requires Internet (Optional Enhancements)

All core features work fully offline. When the Pi has an Ethernet internet connection, three bonus features activate automatically:

- **Live NWS Weather Alerts** — National Weather Service alerts for your area
- **HF Propagation Data** — live solar/geomagnetic data from hamqsl.com
- **APRS-IS Gateway** — internet APRS station data via aprs.fi

System Services & Ports

Incident Management runs several background services, each on its own port. You normally never need to think about these, but they are listed for the net manager and for troubleshooting.

PORT	SERVICE	WHAT IT PROVIDES
80	nginx	Main web server — serves all HTML pages and routes API calls
5050	fcc_lookup_server	Primary API — nets, roster, resources, DMS, FCC lookup, GPS
5051	health_monitor	System health (CPU, memory, disk, temperature, services)
5055	ics_platform_server	ICS incident data — objectives, resources, forms, periods
5056	reference_server	Reference Library — upload, storage, thumbnails
8080	Graywolf	APRS TNC client interface
8081	kiwix-serve	Offline reference library (WikiMed, Wikipedia, iFixit...)
8082	YAAC	Second APRS client (Yet Another APRS Client)
8083	tile_server	Offline map tiles (MBTiles)
8090	Pat (Winlink backup)	Browser-based Winlink backup on the Pi (Express is primary, on the Windows station)
2442	JS8Call	HF digital messaging (requires separate install)

TIP You do not need to memorize ports. Every tool is reachable from the dashboard by tapping its card. Ports matter only when troubleshooting or connecting from a bookmark.

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Getting Started — Connecting to Incident Management

Getting on Incident Management takes three steps: connect to the Wi-Fi, open the dashboard, and identify yourself. The whole process takes under a minute.

Step 1 — Connect to the EMCOMM-NET Wi-Fi

On your phone, tablet, or laptop, open Wi-Fi settings and connect to:

SETTING	VALUE
Network Name (SSID)	EMCOMM-NET
Password	Provided by your net manager (default: fieldcomms2026)
Security	WPA2-PSK
Wi-Fi Channel	6 (2.4 GHz)
Your device receives IP	192.168.50.100 – 192.168.50.200 (automatic)

NOTE The Wi-Fi password is the only access control on Incident Management. Anyone with the password can use every tool — there is no separate login screen. Keep the password posted at the EOC for arriving operators and treat it as an operational credential.

Step 2 — Open the Dashboard

1. Open any web browser (Safari, Chrome, Firefox, Edge — any will work).
2. In the address bar type **http://192.168.50.1** and press Go.
3. The dashboard loads. Bookmark this address for quick access next time.

Friendly aliases also work on most devices: **http://fieldcomms.local** and **http://emcomm.local**.

Step 3 — Identify Yourself

The first time you open any page, an **Identify Yourself** picker appears after about one second. This is covered fully in Chapter 5. In brief: choose ESV Member or Visitor, select your name or enter your IDs, and tap CONFIRM. Served-agency observers can tap **Skip for now**.

What You Can Access

ADDRESS	WHAT IT OPENS
http://192.168.50.1/	Main dashboard and all Incident Management pages
http://192.168.50.1:8081/	Kiwix offline library
http://192.168.50.1:8080/	Graywolf APRS client
http://192.168.50.1:8090/	Pat — browser-based Winlink backup (see Appendix A4)

3 The Main Dashboard

<http://192.168.50.1/>

The dashboard is your central hub. It shows live system status, active weather alerts, a live APRS station table, and quick-launch cards for every tool. At the very top is the ESV organization header with the MCEV/MCEMA logo and your operator identity badge.

Mode Switcher — Amateur Radio vs. ICS

Directly below the header is a two-button mode switcher. It reorganizes the tool cards to match what you are doing. Your choice is saved on your device and remembered.

MODE	WHAT IT EMPHASIZES	WHEN TO USE
■ Amateur Radio	Net Control, APRS, Winlink, Callsign Lookup, Resource Board, Roster, DMS, Pre-Flight, NTS/ICS forms, Propagation, Repeaters, Kiwix, Cheat Sheets	ARES/RACES nets, NTS traffic, weekly nets, exercises
■ ICS / Incident Command	ICS Platform front and center, ICS sections, T-cards, Resource Board, Tactical Map, Roster, Facilities, ICS forms	Declared ICS incidents, multi-agency events

1. Tap ■ **Amateur Radio** or ■ **ICS** in the mode bar.
2. The tool cards rearrange instantly into the chosen grouping.
3. A mode pill in the top-right header shows the current mode. Tap it any time to switch.

Dashboard Sections

SECTION	WHAT IT SHOWS
Hero Bar	Station callsign badge (K9ESV), system name, and a live UTC clock updated every second
NWS Weather Alerts	Live alerts color-coded by severity: EXTREME (red), SEVERE (orange-red), MODERATE (amber), MINOR (blue). Tap any alert to expand full details, onset, expiry countdown, affected counties, and protective actions. Refreshes every 10 minutes when online.
Amateur Radio Mode	Net Control, APRS, Winlink, JS8Call, Callsign Lookup, Tactical Map, HF Propagation, Repeaters, Grid Square, NTS Radiogram, Reference Library, Pre-flight Checklist
Starcom / Public Safety Mode	Starcom Net Logger, Weather Net, SAR Net, Observer Mode, Resource Tracking Map, Resource Board, Member Roster, Facilities Directory, ICS Platform, ICS Forms
ICS Mode	Full ICS Platform (Command, Operations, Planning, Logistics, Finance), ICS Forms, Personnel & Coordination, Communications, Winlink Form Import
APRS Station Table	Live table of APRS stations heard by Graywolf — callsign, distance, course/speed, comment, last-heard. Updates every 30 seconds.
Status Sidebar	System health (CPU, memory, disk, temperature), service status dots (green = running, red = stopped), GPS status, Dead Man's Switch state with armed nets, and an ICS Platform launch button.

TIP Tap any weather alert card to expand it. The expiry countdown updates live so you always know how long an alert remains in effect.

4 Member Roster

<http://192.168.50.1/roster.html>

The Member Roster is the authoritative directory for all MCESV/MCEMA personnel. It tracks every member's identifiers, contact information, certifications, equipment, and deployment activations. It supports both amateur radio operators and non-ham members.

Member Identifiers

Every member is tracked by up to three identifiers. Not all members have all three:

IDENTIFIER	WHO HAS IT	EXAMPLE
ESV Member ID	All ESV members	ESV-042
Starcom Radio ID	All ESV members	1042
Amateur Callsign	Licensed hams only (blank for non-hams)	K9ESV

NOTE Non-ham members are full members of the roster. Leave the Callsign field blank for them — the system uses their Radio ID or Member ID as their primary identifier automatically.

The Two Tabs

TAB	CONTENTS
■ Directory	The full member list as cards. Each shows callsign or member ID, name, roles, phone, email, grid square, certification dots, and equipment badges. Search by name, callsign, or ID at the top.
■ Activations	Members currently checked in to an activation, with role, check-in time, and location. This is where walk-ins promoted from net control appear.

Adding a Member

1. Click **+ Add Member** at the top of the Directory tab.
2. Fill in the fields: **Member ID**, **Callsign** (leave blank if not a ham), **Radio ID**, First Name, Last Name, Phone, Email, and Grid Square.
3. Check any applicable **Roles**: Net Control (NCS), Operator, Liaison, or Emergency Coordinator (EC).
4. Check the member's **Certifications** (ICS-100 through ICS-800, EmComm I/II, CPR/AED, First Aid, CERT).
5. Check their **Equipment** capabilities (HF, VHF, Digital, Packet, PACTOR, VARA HF, VARA FM, APRS, Winlink, Go-Box, Generator, Battery, Vehicle).
6. Click **Save Member**.

Editing or Deleting a Member

1. Click any member card in the Directory to open the edit dialog.
2. Change any field, then click **Save Member**.
3. To remove a member, click **Delete Member** in the edit dialog and confirm.

Member Roles

ROLE	DESCRIPTION
Net Control (NCS)	Primary net controller — manages check-ins and traffic flow
Operator	Standard ARES/RACES field operator
Liaison	Agency liaison — interface between operators and the served agency
Emergency Coordinator (EC)	Oversees ARES/RACES operations for the county

Importing & Exporting (CSV)

Use CSV import to bulk-load members from a spreadsheet, and export for backup or sharing with ARES leadership.

1. To export: click ■ **Export CSV**. A file downloads with all members and all columns.
2. To import: prepare a CSV with these column headers, then click ■ **Import CSV** and choose the file.

Supported columns:

```
member_id, callsign, radio_id, first_name, last_name, role, phone, email, grid, license_class,
roles
```

TIP The **callsign** column may be left blank for non-ham members. The importer accepts both **role** (single) and **roles** (multiple, separated by semicolons), so older roster files still import correctly.

Checking In a Member to an Activation

1. Switch to the **Activations** tab.
2. Type a name or callsign into "Search member to check in..." and select the member.
3. They are added to the active deployment list with a check-in time.
4. To log someone not yet in the roster, click + **Walk-In Check-In**, enter their callsign or radio ID and name, and check them in. Use + **Roster** later to make them permanent.

Exporting the Activation Log

Click **Export Activation Log** on the Activations tab to download a record of who was deployed, their role, and their check-in times — useful for the incident record and after-action reporting.

5 Operator Identity System

Every Incident Management page tracks who is operating at each device so your check-ins, log entries, and form signatures are attributed correctly. There is no password — your identity is stored in your device's browser and remembered between sessions. The system handles all four kinds of people who show up at an activation.

The Four Identity Types

TYPE	IDENTIFIERS USED	BADGE
ESV Member + Ham	Member ID + Callsign + Starcom Radio ID	■ K9ESV
ESV Member, Non-Ham	Member ID + Starcom Radio ID (no callsign)	■ ESV-042
Visitor / Mutual Aid, Ham	Callsign + Agency name	■ W9XYZ
Visitor / Mutual Aid, Non-Ham	Name + Agency (+ optional Radio ID)	■ VISITOR

The system always displays the most useful identifier for radio work. The priority is: callsign first, then Starcom Radio ID, then Member ID, then name.

Identifying Yourself as an ESV Member

1. When the picker appears (or tap your badge in the header to open it), make sure ■ **ESV MEMBER** is selected at the top.
2. Pick your name from the **Select from ESV Roster** dropdown — it is grouped into Amateur Operators and ESV Members. This auto-fills all your fields.
3. Or enter manually: **ESV Member ID**, **Callsign** (leave blank if not a ham), and **Starcom Radio ID**, plus your first and last name.
4. Confirm the preview shows the right person, then tap ✓ **CONFIRM**.

Identifying as a Visitor / Mutual Aid Operator

Operators from other counties or agencies who are not ESV members use the visitor panel.

1. Tap ■ **VISITOR / MUTUAL AID** at the top of the picker.
2. Enter your **First and Last Name** (required).
3. Enter your **Callsign** if you are licensed, and your **Radio ID** if you have been assigned one (both optional).
4. Enter your **Agency / Organization** (required) — for example "Lake County ARES", "Red Cross", "CERT", or "Woodstock Fire Dept".
5. Tap ✓ **CONFIRM**. You will be logged as a visitor with your agency noted.

NOTE Visitors are logged distinctly from ESV members. Net Control can later promote a visitor to the permanent roster — see Chapter 6.

Changing or Clearing Your Identity

1. Tap your identity badge in the top-right of any page header.
2. The picker reopens with your current details. Change them and tap CONFIRM.

Operator Access Cards (Avery 5371)

The net manager can print wallet-sized access cards for every member, generated from the roster. Each card shows the Wi-Fi network and password, the dashboard URL, and the member's identifiers. Ham cards show the callsign large in blue; non-ham cards show the member ID large in purple. All cards show the ESV Member ID and Starcom Radio ID.

To generate cards (run on the Pi):

```
python3 /opt/fieldcomms/scripts/gen_operator_cards.py \ --ssid EMCOMM-NET --password  
YOUR-PASSWORD
```

Or generate from a CSV file instead of the live database:

```
python3 gen_operator_cards.py --csv roster.csv \ --ssid EMCOMM-NET --password YOUR-PASSWORD
```

The output is a print-ready PDF, 10 cards per Avery 5371 sheet. Print, cut, and hand out at activations or meetings so every operator can get on EMCOMM-NET in under a minute.

6 Amateur Net Control Logger

<http://192.168.50.1/netcontrol.html>

The Net Control logger is the primary tool for running ARES/RACES amateur radio nets. It supports multiple simultaneous nets, automatically looks up FCC callsign data, tracks traffic, and exports logs in ICS-309 format.

Starting a Net

1. Click **+ New Net** in the net tabs bar at the top.
2. Enter a **Net Name** — for example "McHenry County ARES Thursday Evening Net".
3. Enter the **Frequency** — for example "147.015 MHz / 107.2 Hz".
4. Enter **Net Control** — your callsign (auto-filled from your identity if set).
5. Click **Activate Net**. The net goes live and appears as a tab.

The Net Selector Tabs

Each active net appears as a tab in the **NET TABS** bar at the top of the page. The active net is highlighted in green. Tap any tab to switch to that net — each net keeps its own separate log, frequency, and net control station. This lets you run several nets at once (for example, a tactical net and a resource net) from one screen.

Logging a Check-In

Each check-in is one row in the log. The local FCC database auto-fills the name and location when you type a callsign — no internet needed.

1. Type the station's **Callsign** in the callsign box. As you type a valid US amateur call, the **Name** and **Location/Grid** fields auto-fill from the FCC database.
2. Set the **Precedence**: ROUTINE, WELFARE, PRIORITY, or EMERGENCY.
3. Add any **Remarks / Traffic** notes (such as an NTS message count).
4. Click **Log Check-In**. The entry is timestamped automatically in UTC and added to the log.

Precedence Levels

PRECEDENCE	COLOR	MEANING
EMERGENCY	Red	Immediate threat to life. Absolute priority over all traffic.
PRIORITY	Amber	Serious urgency — critical supply requests, official needs.
WELFARE	Blue	Health and welfare inquiries to or from a disaster area.
ROUTINE	Normal	Standard traffic, processed in order during normal net cycles.

Walk-Ins and Promoting to the Roster

A station that is not in the FCC database (or whose name you enter manually) is marked with an amber **WALK-IN** badge. You can add any logged station to the permanent roster with one click.

1. Find the entry in the check-in list.
2. Click the green **+ Roster** button on that entry.
3. The station is added to the Member Roster. If they have no ESV number, one is assigned automatically. A confirmation shows the new member ID.

TIP The + Roster button only appears for stations not already in the roster, so you will not create duplicates.

Logging Traffic

1. Switch to the **Traffic** tab.
2. Enter the **From** callsign, **To** callsign or address, traffic type, and a note.
3. Click **Log Traffic**. The traffic item is timestamped and recorded with the net.

The Roster Chips (Quick Check-In)

The **Roster** tab shows quick-tap chips for known stations. Tap a chip to instantly log that station's check-in without retyping the callsign — useful for regulars on a weekly net.

Sharing & Exporting

BUTTON	WHAT IT DOES
■ Observer Link	Copies a read-only URL for the currently selected net. Share it with served-agency staff or EOC viewers — see Chapter 8.
■ ICS-309	Exports the selected net's log as a formatted ICS-309 Communications Log, ready to print or include in an IAP.
■ Backup JSON	Downloads the raw log data as a JSON file for archiving.

Closing a Net

When the net is over, click **Close Net** in the net header. The log is preserved on the server and can still be exported. Export the ICS-309 at the end of each operational period — it is the official communications record.

Drill Mode

Toggle **Drill Mode** in the net header during exercises. A prominent banner marks the session as a drill so logged traffic is never mistaken for a real emergency.

7

Starcom Net Logger

<http://192.168.50.1/starcom.html>

The Starcom Net Logger is its own dedicated module with a dedicated dashboard mode — separate from the Amateur Radio section. Select **Starcom / Public Safety** from the mode bar on the main dashboard to access it. The module is purpose-built for public-safety Starcom radio net logging. It identifies units by Radio ID and unit number rather than amateur callsigns, includes a dispatch center field, and uses the **sc-** prefix convention for net names.

Running Multiple Simultaneous Nets

The Starcom Net Logger supports multiple simultaneous nets — identical to the Amateur Net Control Logger. This is particularly useful during large activations where you may need to run a **Starcom General Net**, a **Weather Net**, and a **SAR Net** all at the same time, each with its own independent check-in log and traffic log.

1. Click **+ New Net** to open the first net.
2. Click **+ New Net** again for each additional net — Weather Net, SAR Net, etc.
3. Each open net appears as a clickable badge in the **active nets panel** on the left sidebar.
4. Click any net badge to switch to that net. The check-in log and traffic log update to show that net's data.
5. Nets run independently — logging a check-in on one net does not affect the others.
6. Close each net individually when it is no longer needed.

TIP Typical multi-activation scenario: Starcom General Net (command channel) + Weather Net (storm spotters) + SAR Net (field teams) all open simultaneously. Net Control switches between them as radio traffic comes in on each channel.

How It Differs from the Amateur Net Logger

FEATURE	AMATEUR LOGGER	STARCOM LOGGER
Unit identification	FCC Callsign (W9XYZ)	Radio ID / Unit number (1234)
Net name prefix	Free form	sc- prefix (sc-McHenry-TAC1)
ID lookup	FCC database auto-fill	Unit name / callsign manual entry
Dispatch field	Not applicable	Dispatch center name field included
Export format	ICS-309	ICS-309 (Starcom header)

Starting a Starcom Net

1. Click **+ New Net**.
2. Enter a net name using the **sc-** prefix, e.g. "sc-McHenry-TAC1".
3. Set the net type, talkgroup, and channel as needed.
4. Click create. The net goes live.

NOTE Always keep the **sc-** prefix on Starcom net names. This is how the system keeps Starcom nets separate from amateur nets in exports and on the Dead Man's Switch monitor.

Setting the Dispatch Center

1. In the entry form, find the **■ Dispatch Center** field at the top.
2. Type the dispatch center name — e.g. "McHenry Co. Sheriff Dispatch".

3. It appears in the net header and stays set for the whole net session, so you do not re-enter it for each check-in.

Logging a Unit Check-In

1. Enter the **Radio ID / Unit #** (e.g. 1234).
2. Enter the **Unit Name / Callsign** (e.g. "MCHENRY CO ARES" or a callsign).
3. Choose the status: Check-In, Traffic, Priority Traffic, Emergency, Dispatch, Check-Out, En Route, On Scene, Available, or Out of Service.
4. Select the talkgroup type and enter the channel/frequency.
5. Add location and remarks, then click **LOG ENTRY**. The time is stamped automatically.

Exporting & Sharing

The Starcom logger shares the same Observer Link, ICS-309 export, and JSON backup tools as the amateur logger (see Chapter 6). Exports carry a Starcom header and keep the sc- prefix.

8

Observer Mode — Read-Only Net View

<http://192.168.50.1/observer.html>

Observer Mode provides a read-only, auto-refreshing view of any active net. It is designed for served-agency personnel, EOC staff, emergency managers, and section leaders who need to monitor radio traffic without touching the net control interface. No login, no identity prompt, no setup required.

What Observer Mode Shows

ELEMENT	DESCRIPTION
Net name & status	Active net name, type (Amateur/Starcom), and open/closed indicator
Live check-in log	All stations or units checked in, with timestamp, location, and remarks
Traffic log	Formal message traffic passed during the net
Station count	Live count of checked-in stations at the top of the page
Auto-refresh	Page refreshes automatically every 15 seconds — no manual reload needed

How to Open Observer Mode

1. On the Net Control Logger or Starcom Logger page, click ■ **Observer Link** to copy the URL to your clipboard.
2. Send the link by email, Winlink, JS8Call, or read it over the air to any operator who needs to monitor.
3. The recipient opens the link in any browser on EMCOMM-NET — phones, tablets, and laptops all work.
4. Observer Mode shows the net immediately with no identity prompt and no login.
5. The net name is encoded in the URL (**observer.html?net=Thursday-Net**), so the link can be bookmarked for recurring nets.

NOTE Observer Mode is completely passive — observers cannot add check-ins, log traffic, or close the net. If an observer needs to interact with the net, they must switch to the Net Control Logger or Starcom Logger page and identify themselves first.

Displaying Observer Mode on a Monitor

Observer Mode works well on a dedicated EOC display or large monitor. Open it in a full-screen browser window and it will continuously show the current net state for everyone in the room. The dark theme and large text are designed to be readable from across a room.

9

FCC Callsign Lookup

<http://192.168.50.1/callsign.html>

The Callsign Lookup page provides instant offline access to the full FCC Amateur Radio ULS database — approximately 800,000 active licensees. No internet connection is required. Lookups are answered in milliseconds from the local SQLite database on the Pi.

Quick Callsign Lookup

1. Type a callsign (e.g. **K9ESV**) in the search box at the top of the page.
2. Press **Enter** or click **Look Up**.
3. The result shows: full name, license class, expiration date, mailing address, grid square, and FRN.
4. Click **+ Add to Roster** to add the licensee directly to the Member Roster.
5. Recent lookups are saved automatically and shown below the search box for quick re-access.

Advanced Search

The Advanced Search tab lets you find callsigns when you don't know the exact call.

1. Click the **Advanced Search** tab.
2. Enter any combination of: First Name, Last Name/Entity, City, State, License Class, or Grid Square.
3. Click **Search**. Results show up to 100 matching licensees.
4. Click any result row to see the full record.
5. Click **+ Roster** on any result to add them to the Member Roster.

Database Status

The FCC database is downloaded during installation from the FCC ULS bulk data files. It is refreshed automatically every Sunday at 03:00 via the **fcc-refresh.timer** systemd timer (requires internet connection). The database status — record count and last update date — is shown at the bottom of the Callsign Lookup page.

NOTE The FCC database covers all active Amateur Radio licensees in the United States. It does not include foreign callsigns, commercial licenses, or expired licenses. For foreign callsigns, use the Reference Library or an internet-connected device.

10 Tactical APRS Map

<http://192.168.50.1/tactical.html>

The Tactical Map is a full-screen Leaflet.js mapping tool that combines live APRS station tracking, offline map tiles, manual tactical overlays, and GPS position tracking. It supports both Graywolf and YAAC as APRS back-ends and works completely offline once map tiles have been downloaded.

Map Layout

ELEMENT	DESCRIPTION
Left sidebar	Station list — all APRS stations received, with callsign, comment, distance, and last heard time
Map area	Interactive Leaflet map with APRS symbols, custom overlays, and drawing tools
Top toolbar	Layer toggles, tile selector, auto-refresh interval, range ring, and drawing tools
Station labels	Callsign labels on the map — toggleable in settings
Status bar	Station count, GPS status, last refresh time

APRS Sources — Graywolf and YAAC

The Tactical Map can receive APRS data from two local back-ends running on the Pi. Both are toggled independently in the toolbar.

SOURCE	PORT	DESCRIPTION
Graywolf	8080	Primary APRS client — receives RF APRS via connected TNC or soundcard interface
YAAC	8082	Secondary APRS client — backup receive and alternate TNC support

Map Tiles and Offline Operation

1. Select a tile set from the **Map Tiles** dropdown in the toolbar.
2. Available tile sets include: USGS Imagery, USGS Topo, OpenStreetMap, Esri Satellite, and 106 geographic presets covering McHenry County and surrounding areas.
3. Tiles are served by the local tile server (port 8083) — no internet needed once downloaded.
4. Download tiles using the **download_tiles.sh** script on the Pi before deployment.

Drawing Tactical Overlays

1. Click a drawing tool in the toolbar: **Marker**, **Polygon**, **Circle**, or **Route**.
2. Click on the map to place the shape. For polygons and routes, click multiple points and double-click to finish.
3. Click a placed marker or shape to add a label or delete it.
4. Overlays are saved to the server automatically and persist across browser sessions.
5. Click **Clear Overlays** to remove all drawn shapes.
6. Click **KML Export** to download the current overlay set as a KML file for use in Google Earth or GIS software.

Station Tracking

1. Click any station in the left sidebar to center the map on that station.
2. Click **■ Track** to lock the map view on a specific station as it moves.
3. Use the **Station age threshold** setting to hide stations older than N minutes.
4. The search box filters the station list by callsign or comment text.

GPS Position

If a USB GPS receiver is connected to the Pi, your current position appears on the map as a blue dot and is used to calculate distances in the station list. The GPS status indicator in the status bar shows the fix quality.

Range Rings

Enter a radius in km in the **Range ring** field and click **Draw Ring** to place a circle on the map centered on your GPS position or the station default coordinates. Useful for visualizing coverage areas.

11

Starcom Resource Tracking Map

<http://192.168.50.1/resmap.html>

The Resource Tracking Map is a purpose-built situational awareness tool for Starcom public safety net operations. It shows unit positions on an offline map with status color coding, supports manual placement and drag-and-drop repositioning, and synchronizes with the Member Roster for unit information. Access it from the Starcom / Public Safety dashboard mode.

Placing a Unit on the Map

1. Click **+ Add Unit** in the left panel.
2. Enter the **Radio ID**, **Unit Name**, and **Type** (choose from the dropdown: Law Enforcement, Fire, EMS, Search & Rescue, Communications, EOC, Shelter, Other).
3. Enter a **Latitude** and **Longitude** directly, or click **■ Pick on Map** to click the map and set the position.
4. Set the **Status** (Available, Deployed, Staging, Out of Service).
5. Click **Save**. The unit appears on the map with its status color.

Status Colors

STATUS	COLOR	MEANING
Available	Green	Unit is on scene and available for assignment
Deployed	Blue	Unit is actively assigned and in the field
Staging	Amber	Unit is staging or en route
Out of Service	Red	Unit is unavailable

Moving Units

1. Click and drag a unit marker on the map to a new position.
2. The unit's coordinates update automatically when you release.
3. Alternatively, click the unit marker, click **Edit**, and update the coordinates manually.

Drawing Zones

1. Click **Draw Zone** to enter zone-drawing mode.
2. Click multiple points on the map to define the zone boundary.
3. Double-click to finish. Enter a zone name and color when prompted.
4. Zones persist across browser sessions and appear on the map for all connected devices.
5. Click **Clear Zones** to remove all zones.

NOTE The Resource Tracking Map and the Tactical APRS Map are separate tools. The Resource Map is for manually-placed Starcom units with known positions. The Tactical Map shows RF-received APRS positions from radio stations automatically.

12

Resource Board

<http://192.168.50.1/resources.html>

The Resource Board tracks all personnel, vehicles, equipment, and facilities assigned to an activation. Each resource is displayed as a card showing its current status. A stats strip at the top of the page shows live counts by status across all resource types.

The Five Status States

STATUS	COLOR	MEANING
Available	Green	On scene and available for assignment
Assigned	Amber	Actively tasked with a specific assignment
Staging	Blue	En route or at the staging area, not yet deployed
Out of Service	Red	Unavailable — mechanical, medical, or administrative hold
Demobilized	Gray	Released from the incident and departed

Adding a Resource

1. Click **+ Add Resource**.
2. Enter the **Name/Description** (e.g. "IC-7300 HF Radio", "Unit 12 — Medic", "Generator — 5kW Honda").
3. Select the **Type**: Personnel, Vehicle, Radio, Generator, Antenna, Computer/Tablet, Medical, Shelter/Tent, Repeater, or Other.
4. Set the **Status** and optionally fill in: Owner/Callsign, Quantity, Location, Assigned To, Assignment/Task, Contact, Notes, and Identifier/Serial #.
5. Click **Save Resource**. The card appears in the board immediately.

Changing a Resource Status

1. Click the colored **status badge** on any resource card.
2. Each click cycles through the five states in order: Available → Assigned → Staging → Out of Service → Demobilized → Available.
3. The stats strip at the top updates instantly.

Editing and Deleting Resources

1. Click the **✎ Edit** button on any card to open the edit form.
2. Update any fields and click **Save Resource**.
3. Click **■ Delete** to permanently remove the resource.
4. Deleted resources are removed from the board and the database immediately.

Filtering the Board

1. Use the **Type** filter dropdown to show only Personnel, Vehicles, Equipment, etc.
2. Use the **Status** filter to show only Available resources or only Assigned resources.

3. Filters can be combined — e.g. show all Assigned Personnel only.
4. The stats strip always reflects all resources regardless of the active filter.

13

Dead Man's Switch — Net Inactivity Monitor

<http://192.168.50.1/deadmans.html>

The Dead Man's Switch (DMS) monitors active radio nets for inactivity. If no check-in or traffic is logged within a configurable threshold, it triggers an audible alert and visual warning. This ensures net control operators don't miss a silent net or a dropped connection during a real activation.

How It Works

When armed for a net, the DMS monitors the net's last activity timestamp. If no check-in, traffic entry, or manual reset occurs within the threshold time, the DMS fires: the page plays an alert sound and displays a countdown warning. The DMS can be set independently for each active net.

Arming the DMS for a Net

1. Open the Dead Man's Switch page from the Amateur Radio or Starcom dashboard.
2. Active nets appear automatically in the panel.
3. Click **■ Arm** next to the net you want to monitor.
4. The DMS is now active. Any check-in or traffic logged in the Net Control Logger or Starcom Logger will automatically reset the timer.
5. The countdown timer is displayed on the page, showing time remaining before the alert fires.

Configuration

SETTING	DEFAULT	DESCRIPTION
Threshold	30 min	Time of inactivity before the alert fires
Warning at	80%	Percentage of threshold at which a yellow warning appears
Poll interval	30 sec	How often the DMS checks the server for net activity
Alert sound	On	Audible alarm when the threshold is exceeded

Resetting and Disarming

1. Click **■ Reset** to manually reset the timer without disarming (use this if you manually checked in all stations without logging it).
2. Click **Disarm** to turn off monitoring for that net.
3. The DMS disarms automatically when the net is closed in the Net Control Logger.

IMPORTANT The DMS runs in the browser tab where you opened it. If you close the tab or navigate away, the audible alert will not sound. Keep the DMS page open in a dedicated tab during activations.

14

Pre-Flight Deployment Checklist

<http://192.168.50.1/preflight.html>

The Pre-Flight Checklist is a structured GO / CAUTION / NO-GO deployment readiness assessment. Before any activation, run through the checklist to verify that all critical systems are operational. The system gives an overall deployment verdict based on the results.

Checklist Sections

SECTION	ITEMS COVERED
Power Systems	Shore/generator power, battery backup/UPS, solar panels, fuel supply, power distribution
Communications Equipment	HF radio, VHF/UHF radio, go-kit radios, HF antenna, VHF antenna, repeater access, Winlink, JS8Call
Computing & Software	Pi server boot, FCC database, nginx, API server, health monitor, paper backup forms
Networking	ASUS router, EMCOMM-NET Wi-Fi, DHCP, field device connectivity, UniFi switch
Field Supplies	Operator cards, printed forms, batteries, cables, tools, maps, personal PPE
Coordination	Net frequency confirmed, EOC contact established, served agency notified, ICS structure assigned

Using the Checklist

1. Open the Pre-Flight Checklist page.
2. Work through each section by clicking the ✓ **Pass** or ✗ **Fail** button for each item.
3. Add a note to any item using the text field — useful for documenting partial issues.
4. Required items marked as Fail will count against the deployment verdict.
5. Optional items can fail without affecting the GO verdict.
6. The verdict banner at the top updates live as you complete items.

Deployment Verdicts

VERDICT	MEANING
■ GO	All required items passed — system is ready for deployment
■ CAUTION	Some optional items failed — review before deploying
■ NO-GO	One or more required items failed — do not deploy until resolved

Exporting the Report

1. Click **Export Report** to generate a printable PDF summary of the checklist.
2. The report includes the verdict, timestamp, and notes for all items.
3. File the report with the incident documentation.

TIP Run the Pre-Flight Checklist before every activation — even for exercises. It takes less than five minutes and catches configuration issues before they become problems during a real emergency.

15 ICS Platform — Overview

<http://192.168.50.1/ics/>

The ICS Platform is the full incident command and documentation system built into FieldComms. It covers all five ICS sections — Command, Operations, Planning, Logistics, and Finance/Admin — plus the ICS Planning P cycle guide. Access it by selecting **ICS** from the mode bar on the main dashboard, or directly at <http://192.168.50.1/ics/>.

Creating a New Incident

1. From the main dashboard (ICS mode), click **■ New Incident** in the active incident strip.
2. Enter the **Incident Name** (required), **Type** (Natural Disaster, HazMat, Search & Rescue, MCI, Weather, Public Health, Other), **Location**, **Jurisdiction**, **Incident Commander**, **Op Period Duration**, and an optional **Situation Summary**.
3. Click **Create Incident**. The incident becomes the active incident for all ICS platform pages.

ICS Navigation

Every ICS platform page has a navigation bar at the top with tabs for each section. The current section is highlighted in amber.

TAB	SECTION	PRIMARY USE
■ Command	Command Section	Incident objectives, safety, weather, situation
■ Operations	Operations Section	T-card board, resource assignments
■ Planning	Planning Section	IAP documents, resource status table, period tracking
■ Logistics	Logistics Section	Comms plan, supply tracking, check-in list, meals
■ Finance	Finance / Admin Section	Cost accounting, time tracking, procurement
■ Planning P	Planning Cycle Guide	Step-by-step ICS planning cycle with forms and attendees

Operational Periods

Each incident has one or more operational periods. The current period number is shown in the incident strip at the top of every ICS page. Click **Advance Period** to move to the next operational period. All ICS data is associated with the incident, not the individual period.

Supported Incident Types

TYPE	TYPICAL USE
Natural Disaster	Severe weather, flooding, earthquake, tornado response
HazMat	Chemical spill, gas leak, radiological incident
Search & Rescue	Lost person, overdue hiker, water rescue
Mass Casualty Incident (MCI)	Multi-patient medical emergency, transportation accident
Weather Net	Starcom/SKYWARN weather spotter activation

TYPE	TYPICAL USE
Public Health	Disease outbreak, mass vaccination, shelter-in-place
Other	Any incident not covered by the above types

16 ICS Command Section

<http://192.168.50.1/ics/command.html>

The Command Section captures the strategic-level information for the incident: incident objectives, safety hazards, weather summary, current situation, accomplishments, and planned actions. This content forms the core of the Incident Action Plan (IAP).

Incident Objectives (ICS-202)

Objectives are the core of the IAP. Every objective should be specific, measurable, action-oriented, realistic, and time-sensitive (SMART).

1. To add a pre-defined objective: click the **Select a common objective** dropdown. Choose from 100 common ICS objectives organized in 13 groups (Life Safety, Communications, Resource Management, Incident Command, Search & Rescue, Shelter & Mass Care, Weather & Natural Disaster, HazMat, Mass Casualty, Public Information, Finance & Administration, Demobilization, MCESV/MCEMA Specific).
2. The selected objective populates the text field below. Edit it if needed — fill in any bracketed placeholders such as [frequency], [location], or [time].
3. If no edits are needed, click **+ Add** or press Enter.
4. To add a completely custom objective: type it directly in the text field and click **+ Add**.
5. Objectives are listed above the input field. Click the **X** next to any objective to delete it.

Safety Message

1. Enter the safety message for this operational period in the **Safety Message** text area.
2. This becomes the ICS-208 Safety Message in the IAP package.

Situation Report Fields

FIELD	WHAT TO ENTER
Current Situation	Brief description of current incident status and key facts
Accomplishments	What has been achieved so far this operational period
Planned Actions	What operations are planned for the next operational period
Weather Summary	Current and forecast weather affecting operations

Saving Command Section Data

1. Click **Save** at the top of the page to save all Command Section data to the server.
2. Data is associated with the active incident and persists across browser sessions.
3. A "✓ Saved" confirmation appears briefly after saving.

17 ICS Operations Section

<http://192.168.50.1/ics/operations.html>

The Operations Section manages the tactical resources assigned to the incident. It uses a T-card style status board where resource cards can be dragged between status columns, mirroring a traditional ICS T-card board.

The T-Card Board

Resources are displayed as cards organized into five columns matching ICS resource status:

COLUMN	MEANING
Available	Resource is checked in and available for assignment
Assigned	Resource has been given a tactical assignment
Out of Service	Resource is unavailable — mechanical, medical, or other
Staging	Resource is at the staging area awaiting assignment
Released	Resource has been demobilized from the incident

Adding a Resource to the Board

1. Click **+ Add Resource** or **+ Assignment**.
2. Enter the resource name, type, callsign/Radio ID, and initial assignment.
3. Click **Save**. The resource appears in the Available column.

Moving Resources Between Columns

1. Drag a resource card from one column to another to update its status.
2. Alternatively, click the resource card to open its detail view, then change the status in the dropdown.
3. All status changes are saved to the server immediately.

Assignment Details

1. Click any resource card to open the detail panel on the right.
2. Enter or update the **Assignment** (Division/Group), **Supervisor**, **Work Location**, and any notes.
3. This information feeds the ICS-204 Assignment List.
4. Click **Save** to commit the changes.

NOTE The Operations Section T-card board and the Resource Board (resources.html) are separate tools. The T-card board is for tactical ICS resource management during an incident. The Resource Board is for general inventory tracking of all equipment and personnel.

18 ICS Planning Section

<http://192.168.50.1/ics/planning.html>

The Planning Section manages the Incident Action Plan documents, resource status summary table, and operational period tracking. It is where the PSC assembles all the pieces of the IAP before each operational period briefing.

IAP Tracker Tab

The IAP Tracker shows which ICS forms have been completed for the current operational period.

1. Click the **IAP Tracker** tab.
2. Each required form (ICS-202 through ICS-208) is listed with a completion checkbox.
3. Check off each form as it is completed and added to the IAP package.
4. The tracker shows the percentage of IAP completion at the top.

Resource Status Summary

1. Click the **Resource Status** tab.
2. The table shows all resources by type with current status counts.
3. Click **+ Add Row** to manually add a resource category.
4. Counts update from the Operations Section T-card board automatically.

Incident Documents

1. Click the **Documents** tab to manage IAP-associated documents.
2. Click **+ Add Document** to link an uploaded reference document to this incident.
3. Enter the document title, form number, and notes.
4. Documents are linked to the incident for archival purposes.

Saving Planning Section Data

1. Click **■ Save** to save all Planning Section data.
2. Data is tied to the active incident and persists across sessions.

NOTE The ICS-309 Communications Log (<http://192.168.50.1/ics309.html>) can be linked to any active incident from its own page. See Chapter 24 for details on the ICS-309 form.

19 ICS Logistics Section

<http://192.168.50.1/ics/logistics.html>

Logistics provides all facilities, services, and materials that support the incident. It owns the communications plan and tracks supplies, food, medical, facilities, and check-in.

The Tabs

TAB	HOW TO USE IT
Comms Plan (ICS-205)	Build the radio communications plan. Each row is a channel: function, channel name, frequency, CTCSS tone, mode, remarks. Click Add Comms Row to add a channel.
Supply	Log supply requests: item, category, quantity needed, quantity on hand, priority. A progress bar shows the fill rate. Use Add Supply .
Facilities	ICP location, base/camp details, staging area manager, and medical aid station info.
Food / Medical (ICS-206)	Meal service log (meal type, count served, menu) plus medical unit leader, ambulance service, and nearest hospital. Use Add Meal .
Check-In (ICS-211)	Personnel check-in log: name/callsign, agency, ICS position, and check-in timestamp. Use Add Check-In .

Building the Comms Plan

1. Open the **Comms Plan** tab.
2. Click **Add Comms Row** and fill in the function (e.g. Command, Tactical 1), channel name, frequency, CTCSS tone, and mode.
3. Repeat for each channel. Click **Save** when done.
4. The Communications Unit Leader and ordering contacts are at the bottom of the tab.

20 ICS Finance / Admin Section

<http://192.168.50.1/ics/finance.html>

Finance/Admin manages all financial aspects of the incident: cost accounting, time tracking, procurement, and administrative records for reimbursement.

The Tabs

TAB	HOW TO USE IT
Cost	Log expenditures: category (personnel, equipment, supply, food, transport), amount, description, vendor, and approver. A running total is shown. Use Add Cost .
Time	Log person-hours per operator: name, position, agency, hours worked, hourly rate (0 = volunteer). Used for reimbursement claims. Use Add Time .
Procurement	Track purchase orders: item, vendor, PO number, amount, status (Requested, Ordered, Delivered, Closed). Use Add Procurement .
Admin	Finance/Admin Section Chief info, billing agency, cost-share type, claim deadline, and supporting documentation requirements.

Logging Volunteer Time

1. Open the **Time** tab and click **Add Time**.
2. Enter the operator name, position, agency, and hours worked.
3. Set the hourly rate to 0 for volunteers (most ESV members).
4. Click save, then **Save** to commit. Export with **CSV** for agency reporting.

21 NTS Radiogram Generator

<http://192.168.50.1/nts.html>

Generate properly formatted ARRL National Traffic System radiograms. The form produces the standard preamble, address block, and text section, and keeps a traffic log.

Creating a Radiogram

1. Click **Auto-fill Number** for the next sequential message number, or enter one.
2. Set the **Precedence**: EMERGENCY, PRIORITY, WELFARE, or ROUTINE.
3. Choose **Handling Instructions** (HX codes) if needed — HXA through HXF.
4. Enter Station of Origin, Place of Origin, and click **Auto-fill Date/Time** for the filed time.
5. Enter the addressee name, address, and phone.
6. Type the message **Text** in ARRL all-caps style. The **Check** (word count) calculates automatically.
7. Enter the **Signature**, then click **Generate Radiogram**.

Handling Instruction (HX) Codes

CODE	MEANING
HXA	Collect landline delivery authorized within (number) miles
HXB	Cancel message if not delivered within (number) hours
HXC	Report date and time of delivery to originating station
HXD	Report identity of station that relayed, plus date/time
HXE	Delivering station to get reply from addressee
HXF	Hold delivery until (date)

Saving and Sending

1. Click **Save to Log** to store the radiogram in the traffic log.
2. Click **Copy Text** to copy the formatted radiogram to the clipboard.
3. Click **Send to Net** to push it toward the active net for passing.
4. Reload any past radiogram from the log with **Load** on its log entry.

22 ICS-213 General Message

<http://192.168.50.1/ics213.html>

The ICS-213 is the standard written message form for inter- and intra-agency communications on an incident. Use it when you need a written record of a request, situation update, or resource order.

Completing an ICS-213

1. Fill in the header: incident name, To, From, position/agency, date/time, and subject.
2. Click **Auto-fill** to populate the current date and time.
3. Type the message in the body.
4. Click **Generate Form** to render a print-formatted version.
5. Use the **Print** button to produce a field-ready paper copy.
6. Click **Save to Log** to keep a copy in the form log.

NOTE Use the ICS-213 for formal written traffic. For routine spoken radio traffic, use the Net Control log instead.

23

ICS-214 Activity Log

<http://192.168.50.1/ics214.html>

The ICS-214 is a unit-level activity log required for every ICS section and unit. It records personnel assigned and a timestamped log of activities through the operational period.

Completing an ICS-214

1. Enter the incident name, operational period, unit name, and unit leader.
2. Click **Add Personnel** to list each person assigned to the unit (name, position, agency).
3. For each activity, click **Add Entry**, then **Auto Timestamp** for the current time, and type what happened.
4. Entries accumulate in time order throughout the period.
5. Click **Generate Form** and **Print** for the incident record. **Save Local** keeps a copy on the device.

ICS-309 Communications Log

The companion ICS-309 page (Dashboard → ICS mode → ICS-309 Comms Log) records a log of message traffic — required for after-action reports. Enter the incident, operational period, operator, and station/net in the header, then add a row per message with its date/time, who it was from, who it was to, and the subject. Use **Add (timestamp now)** to stamp the current UTC time on a new row. Click **Generate Form** for the printable ICS-309, or **Save to Incident** to file it in the incident record. The Net Control and Starcom loggers can also export their net logs directly as an ICS-309.

TIP Communications logs are required for after-action reporting. The ICS-309 page, the net loggers, and the Winlink Form Import (next chapter) all produce ICS-309 records that land in the same incident archive.

24 Winlink Form Import & Incident Archiving

<http://192.168.50.1/winlink-import.html>

Winlink Express has its own built-in ICS forms (ICS-213, ICS-214, and many others). Operators fill them out in a browser window that Express opens, then Express transmits the data. This chapter explains how to bring that form data into the Incident Management server so the whole incident is documented and archived in one place — and how to re-print a received ICS-213 or ICS-214 on the Pi.

How Winlink Forms Work (Background)

When you choose a form template in Winlink Express, it opens an HTML form in your web browser. After you fill it in and click Submit, Express does two things: it writes a readable summary into the message body, and it attaches a small data file named `RMS_Express_Form_*.xml` that contains the field values. On the receiving Express, that XML is merged back with the matching template to re-draw the form for printing. The pretty re-rendering is a Winlink Express feature — the Pi does not run Express's forms engine — but the XML carries all the data, and the Incident Management server uses it to capture, archive, and (for ICS-213/214/309) re-print the form.

Getting a Form Into the Server

Open **Winlink Form Import** from the dashboard (ICS mode). Then:

1. In Winlink Express, open the form message (sent or received).
2. Either **save the attached** `RMS_Express_Form_*.xml` file (right-click the attachment → Save), or **copy the message text**.
3. On the Winlink Form Import page, drag the saved XML file onto the drop zone (or click to browse), or paste the XML / message text into the box.
4. Click **Parse Form Data**. The page detects the form type and extracts the fields.

TIP If you only have the message text and not the XML attachment, paste the text — the importer will still pull out the labeled fields. The XML attachment gives the cleanest result, so save it when you can.

Review and Correct

The importer shows every field it extracted in an editable grid. Check them against the original — Winlink form versions change over time, so occasionally a field lands in the wrong box or under **Fields not automatically mapped**. Fix anything that looks off before filing. Unmapped fields are still preserved in the archived record, so nothing is lost.

Tag to the Incident and File

1. Choose the **Incident** this form belongs to (the dropdown lists active incidents from the ICS Platform).
2. Choose the **Direction** — Received (incoming) or Sent (outgoing).
3. Click **Open in ICS form (print)** to re-render an ICS-213 or ICS-214 on the Pi, ready to print, or
4. Click **Archive to incident** to file the form data into the incident record without printing.

NOTE Archived forms are stored on the server tagged with the incident, so they become part of the incident's permanent record alongside the forms generated directly on the Pi. If the server is briefly unreachable, the importer keeps a local copy on the device so the data is not lost — re-archive it once the server is back.

What Re-Rendering Supports

FORM	RE-PRINT ON PI?	NOTES
ICS-213 General Message	Yes	Re-renders as the Pi's ICS-213 for printing
ICS-214 Activity Log	Yes	Activity and personnel lines are split into rows automatically
ICS-309 Communications Log	Yes	Log rows (date/time, from, to, subject) are parsed into the comms log
Other Winlink forms	Archive only	Data is captured and archived; ICS re-print is for 213/214/309

For forms other than ICS-213/214/309, the data is still fully captured and archived to the incident — you simply do not get an ICS-formatted re-print. For those, keep the Winlink Express printout or PDF as the formatted copy and use the archive for the data record.

Recommended Workflow

1. Run Winlink traffic in Express as your operators normally do.
2. At the end of each operational period (or as forms come in), import the ICS forms into the server and tag them to the incident.
3. For ICS-213/214/309, re-print on the Pi if a paper copy is needed for the IAP or the served agency.
4. When the incident closes, all forms — those created on the Pi and those imported from Winlink — are together in the incident record for archiving and after-action review.

25

HF Propagation

<http://192.168.50.1/propagation.html>

The propagation page fetches live solar and geomagnetic data (from hamqsl.com, when internet is available) and uses it to estimate current HF band conditions.

Reading the Indicators

INDICATOR	WHAT IT MEANS FOR HF
SFI (Solar Flux)	Higher = better high-band propagation. >130 excellent; 90–130 good; <90 40m/80m only.
Sunspot Number	More sunspots = more solar activity = better HF. Follows the 11-year cycle.
A-Index	Daily geomagnetic activity. <10 quiet (good HF); 10–30 unsettled; >30 disturbed.
K-Index (0–9)	3-hour geomagnetic snapshot. ≤2 quiet; 3–4 unsettled; ≥5 storm (HF degraded); ≥7 severe.
X-Ray Flux Class	Solar flare class. A/B/C minor; M moderate (SID possible); X strong HF blackout risk.
Band Condition Bars	Green good / Amber fair / Red poor, for each band, day and night.
MUF / LUF	Max and min usable frequencies for local, regional, DX, and long-path circuits.

Using It

Click **Fetch / Refresh** to pull the latest data. It refreshes every 15 minutes when online. With no internet, the page shows the last known values.

TIP For EmComm: 40m is your backbone band — reliable day and night for regional to national paths. 80m excels at night for local/regional. The 60m channels (5330.5–5403.5 kHz USB) are the interoperability standard for served-agency communication. Monitor 14.300 MHz for IARU distress.

26 Repeater Database

<http://192.168.50.1/repeaters.html>

The Repeater Database displays repeater data with rich filtering. It works in three modes: an offline file you load yourself (the most reliable method), the server API, or built-in sample data. Real repeater data comes from RepeaterBook.com.

The Three Sources

SOURCE	HOW TO USE IT
Offline File (recommended)	Load a RepeaterBook CSV or JSON export. Drag-and-drop the file onto the drop zone, or click Browse. Data persists in your browser after the first load. This needs no API token and is the most dependable way to get real data onto the system.
Server API	Pulls from the Incident Management server if the net manager has run <code>fetch_repeaters.py</code> to download RepeaterBook data. This now requires a RepeaterBook API token (see below). Click Refresh to reload.
Sample Data	A few clearly-labeled PLACEHOLDER entries (Callsign "SAMPLE-1", etc.) that show how the page looks before you load real data. These are NOT real repeaters and must never be used for actual operations.

IMPORTANT The built-in sample entries are placeholders only — fictional callsigns and frequencies used to demonstrate the page layout. Always load real RepeaterBook data before an activation. Do not rely on the sample data for any on-air operation.

Loading Real Data — Server API Method (Requires Token)

This is the primary method for MCESV/MCEMA: the net manager fetches data from RepeaterBook into the Incident Management database, and it is then served to every device on the Server API tab. Because repeater data changes infrequently, this runs automatically once a month, and can also be run on demand. As of 2026, RepeaterBook requires an approved API token.

One-Time Token Setup

1. Create a RepeaterBook.com account and apply for API access at repeaterbook.com/api/token_request.php.
2. Once approved, save the token on the Pi. Either create `/opt/fieldcomms/data/repeaterbook_token.txt` containing just the token, or create `/opt/fieldcomms/data/repeaterbook.env` with the line `REPEATERBOOK_TOKEN=your-token` (the monthly auto-refresh reads this env file).

Fetching On Demand

Run the update tool and choose **Fetch repeater data** (option 6) — it will prompt for the token the first time and then for the states to fetch. Or run the fetcher directly:

```
export REPEATERBOOK_TOKEN="your-issued-token" python3 /opt/fieldcomms/python/fetch_repeaters.py \
  --states IL,WI,IN,IA --bands 2m,70cm
```

The fetcher writes the repeaters directly into the Incident Management database (replacing the previous RepeaterBook-sourced rows, while leaving any manually-added rows intact) and also saves a JSON backup. On the Repeater Database page, select the **Server API** tab and click **Refresh** to display the data.

Automatic Monthly Refresh

A systemd timer (repeater-refresh.timer) runs the fetch automatically on the first of each month at 04:00, as long as the token is saved in the repeaterbook.env file and the Pi has internet at that time. Check or trigger it manually:

```
systemctl status repeater-refresh.timer # next scheduled run  
sudo systemctl start repeater-refresh.service # run now  
journalctl -u repeater-refresh -n 30 # view last run log
```

TIP The default region is Illinois plus the bordering states (IL, WI, IN, IA) on 2m and 70cm — appropriate for McHenry County and mutual-aid areas. Edit the --states list in repeater-refresh.service to change coverage.

Offline File Method (No Token / No Internet on the Pi)

If you do not have an API token, or the Pi has no internet, you can still load real data manually from a RepeaterBook CSV export:

1. On any device with internet, go to repeaterbook.com and search your area.
2. Use Export → CSV.
3. On the Repeater Database page, select the **Offline File** tab and drag-and-drop the CSV (or click Browse). The data persists in that browser.

Filtering and Sorting

Once real data is loaded, filter by Band (10m through 23cm), Mode (FM, D-STAR, C4FM-Fusion, DMR, P25), Status (On-Air/Off-Air), State, Use (Open/Closed/Private), and EmComm role (ARES/SKYWARN/RACES). Click any column heading to sort; click distance to sort nearest-first from your station coordinates.

Viewing a Repeater

1. Click any repeater row to open its detail panel.
2. Use **Copy** to copy its details, or **Program** to view programming info.
3. Export the filtered list with **Export CSV**.

27 Facilities Directory

<http://192.168.50.1/facilities.html>

Maintain a directory of EOC locations, hospitals, shelters, staging areas, and command posts. Each entry stores address, coordinates, radio frequencies, CTCSS tone, contact person, on-site callsign, generator status, ADA access, capacity, and operational notes.

Default Facilities

On first startup, Incident Management seeds four McHenry County facilities: the McHenry County EOC (Woodstock), Centegra Hospital Woodstock, the McHenry County Fairgrounds staging area, and Centegra Hospital McHenry. Edit these with your actual operational details.

Managing Facilities

1. Click **+ Add** to create a facility, or click any facility card to view its full detail.
2. In the detail view, click **Edit** to change fields, or **Delete** to remove it.
3. Use **Copy Address** to copy a facility's address to the clipboard.
4. Use **Export CSV** to back up the directory.

TIP Update the Facilities page before every deployment season so EOC, shelter, and hospital frequencies and contacts stay current.

28 Grid Square Calculator

<http://192.168.50.1/grid.html>

Convert between decimal latitude/longitude and Maidenhead grid squares in both directions, and calculate distance and bearing between two grid squares. Supports 4-character (EN52) and 6-character (EN52ab) precision.

Using the Calculator

1. To convert coordinates to a grid square: enter latitude and longitude, click **Calculate from Lat/Lon**.
2. To convert a grid square to coordinates: enter the grid square, click **Calculate from Grid**.
3. To use your current location: click **Use My Location** (requires location permission, or uses the configured/GPS coordinates).
4. To find distance and bearing: enter two grid squares and click **Calculate Distance**.

The map on the page highlights the grid square on a North America outline for quick visual reference.

29 Radio Cheat Sheets

<http://192.168.50.1/cheatsheets.html>

A quick-reference set of radio and ICS cheat sheets, organized into tabs. No internet needed — everything is built in.

TAB	CONTENTS
Phonetic Alphabet	NATO phonetics A–Z with pronunciation, plus ITU number pronunciation (Zero, WUN, TOO, TREE, FOW-er, FIFE, SIX, SEV-en, AIT, NIN-er)
Q-Codes	Common amateur Q-codes (QRM, QRN, QRP, QSL, QTH...) and EmComm net Q-codes (QNI, QNS, QND, QNN...)
Prowords	Full procedure word reference — ROGER, WILCO, SAY AGAIN, BREAK, OVER, OUT, CORRECTION, SILENCE, AUTHENTICATE, and more, with meanings and examples
Band Plan	2m and 70cm segments, HF emergency frequencies (ARES, ARRL, IARU, 60m channels), and service bands (MURS, FRS, GMRS, Marine, Aviation)
NTS Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE definitions and ICS triage colors (Immediate / Delayed / Minor / Expectant)
ICS Structure	ICS org chart and the key ICS forms reference table (ICS-201 through ICS-309)
CTCSS / DCS	Full CTCSS tone table (67.0–250.3 Hz) and common DCS codes
Signal Reports	RST scale, APCO P25 BER quality, FM voice report shorthand, and dBm-to-watts

Tap any tab to view that reference. These are designed to be readable on a phone in the field.

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Print Center

<http://192.168.50.1/printcenter.html>

The Print Center is a one-stop hub for every printable document, plus a built-in incident cover-sheet generator.

CATEGORY	DOCUMENTS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, NTS Radiogram, Pre-Flight Checklist
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, CTCSS/DCS & Signal Reports
Operations	Net Control Log (ICS-309), Starcom Net Log (ICS-309), Member Roster, Resource Board
Cover Sheet	Generate a formatted IAP cover sheet from incident details

Generating an Incident Cover Sheet

1. Fill in incident name, number, IC, operational period, frequency, location, and situation summary.
2. Click **Generate Cover** to preview it in the page.
3. Click **Print Cover** to print it in a new window.
4. Click **Clear** to reset the form.

For any other document, click its card to open a print preview, then print from your browser. ICS-309 logs are exported from Net Control / Starcom (their pages open via the links here).

Connecting a Printer to EMCOMM-NET

FieldComms has print buttons on 17 pages. All printing uses the browser standard print function — it sends the job to whatever printer the operator's browser can reach. Three options are supported:

OPTION	HOW IT WORKS	BEST FOR
A — Client's own printer	Each operator prints to their own connected printer. No Pi setup needed.	Single operator with their own printer
B — USB printer via Pi (CUPS)	USB printer on the Pi shared across EMCOMM-NET via CUPS print server. Installed automatically. Admin at http://192.168.50.1:631 . Supports Windows, Mac, iOS AirPrint, Android, Chromebook.	Field activations — one shared printer for the whole team
C — Network printer	Wi-Fi or Ethernet printer on EMCOMM-NET directly. Devices discover it via Bonjour/mDNS automatically.	Sites with an existing Wi-Fi capable printer

Setting Up Option B — USB Printer via CUPS

1. Plug the USB printer into the Pi or powered USB hub.
2. Open <http://192.168.50.1:631> from any device on EMCOMM-NET.
3. Click **Administration** → **Add Printer** and log in with the Pi username/password.
4. Select the printer from Local Printers. Check **Share This Printer**. Click Continue.
5. Select the driver, click Add Printer, then print a test page.
6. Windows: Settings → Printers → Add a printer — it appears automatically on EMCOMM-NET.
7. iOS/iPad: tap Share → Print in any FieldComms page — AirPrint finds it automatically.

8. Android: install CUPS Print app → add printer at 192.168.50.1 port 631.

TIP Recommended field printers: **Brother HL-L2350DW** (laser, excellent Linux support), **Canon PIXMA TR150** or **HP OfficeJet 200** (battery-powered portable options). See the Installation Guide Step 8 for full setup instructions and the complete comparison of all three printer connection options.

31 Reference Library

<http://192.168.50.1/refs.html>

The Reference Library is a document-management system for all field reference materials — radio manuals, interoperability plans, ICS form packages, SOGs, agency plans, and training documents. Files are stored on the Pi and accessible from any device on EMCOMM-NET.

The Three Tabs

TAB	CONTENTS
■ Amateur Radio	Radio manuals, frequency plans, ARRL publications, band plans, antenna refs
■ ICS / Emergency Mgmt	SWIC plans, NIMS docs, ICS form packages, agency SOGs, mutual aid plans
■ All Documents	Every uploaded document regardless of category

Uploading a Document

1. Click **■ Upload Document** — a panel slides in from the right.
2. Drag-and-drop a file or click Browse. Accepts PDF, Word, Excel, PowerPoint, images, KML, ZIP, GPX — up to 200 MB.
3. Fill in Title, Category, Source/Publisher, Applies To, Description, Tags, Revision, and an Expiry date if relevant.
4. Under **Show on tab(s)**, check Amateur Radio, ICS/EmMgmt, or both. Checking both makes the document appear on both tabs (cross-referencing) without storing two copies.
5. Click **■ Upload Document**. PDFs get an automatic thumbnail.

Finding and Using Documents

1. Use the left sidebar to filter by **Category** (Radio Manual, ICS Form/Plan, Agency, Training, Plan/Procedure, Other) or by **Cross-Referenced**.
2. Click any **tag** in the tag cloud to filter by topic.
3. Sort by Newest, Title, Source, or Most Downloaded. Switch between grid and list view.
4. Click a document to open its detail; use **Download** to open it, **Edit** to change its metadata, or delete it.

TIP Ten ICS position job-aid PDFs (IC, Safety Officer, PIO, Liaison, the four Section Chiefs, Net Control Station, and Field Operator) have been pre-generated with ESV branding and are ready to upload to the ICS tab.

32 Kiwix Offline Library

<http://192.168.50.1:8081/>

Kiwix serves a complete offline reference library from the Pi on port 8081. Every device on EMCOMM-NET can browse it with no internet. Content is stored as ZIM files (compressed, self-contained web archives) organized into three tiers by size and operational value.

Accessing Kiwix

Open a browser to **<http://192.168.50.1:8081>**, or tap the **Kiwix Library** card on the dashboard. Kiwix has a built-in full-text search bar that searches across all installed content at once — type any symptom, procedure, or topic and results appear instantly, offline.

Content Catalogue

TIER	CONTENT	SIZE	BEST FOR
1	WikiMed Medical Encyclopedia	~471 MB	Mass-casualty events, medical support
1	Wikipedia (Mini)	~1.2 GB	General field reference, briefings
1	Wikivoyage	~820 MB	Shelter planning, evacuation routes
2	Wikibooks (field manuals)	~4.2 GB	First aid, survival, ham radio, field skills
2	iFixit Repair Manuals	~3.1 GB	Equipment repair without internet
3	Wikipedia Medicine (full)	~8.5 GB	Advanced clinical reference
3	Wikiversity	~3.4 GB	ICS/CERT training, education
3	Electronics Stack Exchange	~2.2 GB	Radio circuit & antenna troubleshooting

INSTALL SET	TOTAL SIZE	STORAGE NEEDED
Tier 1 (Essential)	~2.5 GB	Fits any 8 GB Pi SD card
Tiers 1+2	~10 GB	16 GB SD or USB drive
All tiers	~22 GB	32 GB SD or USB drive

Installing Content (Net Manager)

ZIM files are installed with the `kiwix_setup.sh` script on the Pi. You must have internet to download — do this before deployment, not during an activation.

COMMAND	WHAT IT DOES
<code>sudo bash kiwix_setup.sh</code>	Interactive menu — pick individual ZIMs or a tier
<code>sudo bash kiwix_setup.sh --tier 1</code>	Install all Tier 1 content (~2.5 GB)
<code>sudo bash kiwix_setup.sh --tier 2</code>	Install Tier 1 + 2 (~10 GB) — recommended
<code>sudo bash kiwix_setup.sh --tier 3</code>	Install all content (~22 GB)
<code>sudo bash kiwix_setup.sh --list</code>	Show all ZIMs with size, tier, install status

COMMAND	WHAT IT DOES
<code>sudo bash kiwix_setup.sh --status</code>	Show installed ZIMs and service status
<code>sudo bash kiwix_setup.sh --update</code>	Check for and download newer ZIM versions

Adding a Custom ZIM

1. Download a ZIM from kiwix.org to a device with internet.
2. Copy it to the Pi: `scp mybook.zim fieldcomms@192.168.50.1:/opt/kiwix/zim/`
3. Register it: `sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/mybook.zim`
4. Restart: `sudo systemctl restart kiwix`
5. Open <http://192.168.50.1:8081> — the new content appears.

File Locations & Service

ITEM	PATH / COMMAND
ZIM files	<code>/opt/kiwix/zim/</code>
Library index	<code>/opt/kiwix/library.xml</code>
Service status	<code>sudo systemctl status kiwix</code>
Restart service	<code>sudo systemctl restart kiwix</code>
View log	<code>sudo journalctl -u kiwix -f</code>

TIP Kiwix is a separate service from Incident Management (kiwix.service on port 8081). Its green/red status dot appears in the dashboard Health Monitor sidebar. Install WikiMed (Tier 1) at minimum — it is the most critical content for EmComm and only ~471 MB.

33 Offline Maps, GPS & Health Monitor

This chapter covers three infrastructure features: the offline map tile system, GPS live position, and the system Health Monitor.

32.1 Offline Map Tile System

Map tiles are downloaded in advance, stored as MBTiles files on the Pi, and served locally on port 8083. Both the Tactical APRS Map and Starcom Resource Map use offline tiles by default. The default source is **USGS Imagery+Topo Hybrid** — public-domain satellite imagery with roads, contours, and place names.

Downloading Tiles (Net Manager)

1. Run `sudo bash /opt/fieldcomms/scripts/download_tiles.sh` for the interactive menu (106 presets: all Illinois counties, WI/IN/IA border counties, all 50 states).
2. Search presets: `download_tiles.sh --search "dupage"`
3. Download a preset: `download_tiles.sh --area county_mchenry`
4. Custom area: `--north 42.70 --south 41.90 --west -89.00 --east -87.80 --zoom 8-16`
5. Downloads are resumable — re-run an interrupted download and it continues.

ZOOM	DETAIL	BEST FOR
8–10	State/county overview	Incident context, mutual aid area
11–13	Township/city	Sector assignments, shelter locations
14	Street-level	Field operator navigation
15–16	Block/building	SAR grids, door-to-door

McHenry County recommended: Essential Z8–14 (~8 MB), Standard Z8–16 (~60 MB), Full Z8–17 (~550 MB).

32.2 GPS Live Position

A USB GPS receiver gives Incident Management a live station position. When a fix is acquired, the tactical map marker, NWS weather alert zone, and stored coordinates all update automatically.

FEATURE	WITH GPS FIX	WITHOUT GPS
Tactical Map	Marker moves to live position every 30 s	Marker at configured coordinates
Weather Alerts	Alerts for your actual location	Alerts for configured county
Dashboard Sidebar	Live lat/lon and fix quality	Shows "No Fix" + configured position

Supported hardware: USB GPS receivers with CP2102, CH340, u-blox NEO-6/7/8, Adafruit Ultimate GPS, PL2303, or FTDI chipsets. Auto-detected by udev rules and exposed as `/dev/gps0`. No GPS is fine — coordinates are set at install time.

Setting Fixed Coordinates (No GPS)

```
curl -X POST http://192.168.50.1:5050/api/station/config \ -d  
'{"lat":42.3247,"lon":-88.3822,"callsign":"K9ESV"}'
```

32.3 Health Monitor

The health monitor runs as a background service on port 5051 and feeds the dashboard sidebar with live system diagnostics. The raw data is at <http://192.168.50.1:5051/health>.

METRIC	NORMAL	ACTION IF EXCEEDED
CPU Temperature	< 70°C	Improve ventilation. Pi 5 throttles at 85°C.
Memory Usage	< 80%	Close unused browser tabs; restart non-essential services
Disk Usage	< 90%	Archive old logs; remove unused ZIM files
Service Status	All green	Red dot = service down. Restart with systemctl.
Internet	Connected (if avail.)	Check Ethernet. Core features work offline.
GPS	Fix (if attached)	No GPS is fine — coordinates configured at install.

Restarting a Stopped Service

```
sudo systemctl restart fcc-lookup # or: health-monitor, # ics-platform, fieldcomms-refs, kiwix, deadmans
```


34

Network Hardware — ASUS RT-BE58 Go & UniFi Switch Flex 2.5G

FieldComms v1.0 uses a dedicated Wi-Fi 7 travel router and a 2.5 GbE managed switch instead of the Pi broadcasting its own Wi-Fi hotspot. This gives every device on scene a faster, more reliable connection, lets the Pi focus entirely on serving the FieldComms application, and provides wired 2.5 GbE ports for latency-sensitive devices like the radio TNC.

Network Architecture Overview

DEVICE	ROLE	CONNECTION
ASUS RT-BE58 Go	Wi-Fi 7 access point + DHCP server + WAN gateway	WAN: Ethernet uplink or USB tether · LAN: 2.5G to UniFi switch
UniFi Switch Flex 2.5G-5	5-port 2.5 GbE wired distribution switch	Port 1 (PoE in): uplink to ASUS router · Ports 2–5: Pi, TNC, EOC laptop, spare
Raspberry Pi 5 (16 GB)	FieldComms application server	Wired 2.5 GbE to UniFi switch — assigned static IP 192.168.50.1
On-scene devices	Operator browsers, tablets, phones	Wi-Fi (EMCOMM-NET, 2.4 GHz or 5 GHz) — DHCP 192.168.50.100–200
TNC / Radio interface	Graywolf / YAAC APRS	Wired or USB direct to Pi — no network required for TNC itself

NOTE The Pi no longer runs `hostapd` or `dnsmasq`. The ASUS router owns the Wi-Fi network and DHCP. The Pi is a plain wired client with a static IP. This is configured during install — see Section 33.5 below.

33.1 ASUS RT-BE58 Go — Initial Setup

The RT-BE58 Go ships in Router mode. For FieldComms you keep it in Router mode so it handles DHCP, NAT, and Wi-Fi for all on-scene devices.

- 1. Power:** Connect the included USB-C cable (18W PD) to a power bank or AC adapter. The router draws under 10W at full load — any USB-C PD source rated 18W+ works.
- 2. First boot:** Connect a laptop to the ASUS router via Ethernet (or to its default Wi-Fi SSID printed on the label). Open a browser to **`http://192.168.1.1`**.
- 3. Quick Internet Setup wizard:** Step through the wizard. Select **Router** mode when prompted. For WAN source choose **Automatic IP** if you have an Ethernet uplink, or skip WAN setup entirely for a fully offline deployment.
- 4. Set the LAN subnet:** Go to **LAN** → **LAN IP**. Change the router's LAN IP to **192.168.50.1** and the subnet mask to **255.255.255.0**. Click Apply. The router will reboot. Reconnect to **`http://192.168.50.1`** after reboot.
- 5. DHCP range:** Go to **LAN** → **DHCP Server**. Set the IP pool to **192.168.50.100 – 192.168.50.200**. This reserves the .1–.99 range for static assignments (Pi is .1, other fixed devices can use .2–.99).
- 6. Add a DHCP reservation for the Pi:** Go to **LAN** → **DHCP Server** → **Manually Assigned IP**. Enter the Pi's MAC address and assign it **192.168.50.1**. Alternatively configure a static IP on the Pi itself (Section 33.5).

7. Set the Wi-Fi SSID: Go to **Wireless** → **General**. Set both the 2.4 GHz and 5 GHz SSIDs to **EMCOMM-NET**. Set a strong WPA3/WPA2 password. Apply.

8. Disable Wi-Fi password prompt (optional): For operations where the router is physically secured at the EOC, some teams use an open network for EMMCOMM-NET so any operator device connects without a password. Only do this if physical access is controlled.

IMPORTANT Write down the ASUS router admin password and store it on a USB drive labeled FIELDCOMMS. If the router is factory-reset in the field you will need it to reconfigure the LAN subnet and SSID before FieldComms is accessible.

33.2 ASUS RT-BE58 Go — WAN / Internet Options

The RT-BE58 Go supports several WAN sources. For EmComm you can use any of these depending on whether internet is available at the incident location.

WAN SOURCE	HOW TO CONFIGURE	WHEN TO USE
No WAN (offline only)	Skip WAN setup or leave WAN unplugged. All FieldComms features work. NWS weather alerts, HF propagation data, and APRS-IS will not function.	Field deployments with no internet — default EmComm posture
Ethernet uplink (EOC / shelter)	Plug the site's Ethernet into the ASUS WAN port. Set WAN type to Automatic IP.	EOC with existing network — enables bonus internet features
USB smartphone tether	Enable USB tethering on your phone, connect to ASUS USB port. Set WAN type to USB.	Field site with cellular coverage — Starlink or 5G hotspot
WISP (connect to existing Wi-Fi)	In the ASUS web UI select WISP mode. The router connects upstream via Wi-Fi and rebroadcasts as EMMCOMM-NET.	Hospital / shelter with guest Wi-Fi available

33.3 UniFi Switch Flex 2.5G-5 — Setup

The USW-Flex-2.5G-5 is a compact 5-port 2.5 GbE switch. It requires no software configuration for basic FieldComms use — it works as an unmanaged switch out of the box and is managed via the UniFi Network app when connected to a UniFi console.

Physical Setup

- 1. Power:** The switch is powered via the included 5V/1A USB-C adapter, or by PoE input on port 1 if your uplink device provides PoE. For most deployments use the USB-C adapter connected to a USB power bank or the ASUS router's USB port.
- 2. Uplink:** Connect Port 1 (the PoE-in / uplink port) to the ASUS router's 2.5G LAN port using a Cat 6 or better cable.
- 3. Pi 5:** Connect Port 2 to the Raspberry Pi 5's Ethernet port (2.5G capable).
- 4. Additional wired devices:** Ports 3–5 are available for an EOC laptop, a second radio TNC, a Starlink terminal, or other fixed devices.

Port Assignment Reference

PORT	SPEED	ASSIGNMENT
Port 1	2.5 GbE (PoE in / uplink)	ASUS RT-BE58 Go — LAN port

PORT	SPEED	ASSIGNMENT
Port 2	2.5 GbE	Raspberry Pi 5 — FieldComms server
Port 3	2.5 GbE	EOC laptop / Net Manager workstation (optional)
Port 4	2.5 GbE	Spare — second TNC, Starlink router, or IP radio
Port 5	2.5 GbE	Spare

TIP The USW-Flex-2.5G-5 can be managed via the UniFi Network app for VLAN assignment, port isolation, and traffic statistics, but this requires a UniFi console (Cloud Key, Dream Machine, or self-hosted UniFi controller). For basic FieldComms operation none of that is required — plug it in and it works.

33.4 UniFi App Setup (Optional — Advanced)

If you want to use the UniFi Network app to manage the switch (VLANs, port stats, firmware updates), install the UniFi Network application on the EOC laptop or the Pi itself.

1. Install UniFi Network on the EOC laptop: download from ui.com/download/unifi.
2. On first launch, create a local-only UniFi site (no cloud account required for local use).
3. The app will discover the USW-Flex-2.5G-5 automatically if it is on the same subnet.
4. Adopt the switch into your local UniFi site.
5. Optional: enable port isolation on ports 4–5 if you want to prevent spare-port devices from accessing FieldComms directly.

NOTE Do not connect the switch to the UniFi cloud if operational security is a concern. The switch works at full speed with no cloud connection.

33.5 Raspberry Pi 5 — Static IP Configuration

The Pi must always be reachable at **192.168.50.1**. Configure a static IP via NetworkManager (the default network manager on Raspberry Pi OS Bookworm).

Run these commands on the Pi (over SSH or with a keyboard attached):

```
# Find the Ethernet interface name (usually eth0) ip link show # Set a static IP via
NetworkManager sudo nmcli con mod "Wired connection 1" \ ipv4.addresses 192.168.50.1/24 \
ipv4.gateway 192.168.50.1 \ ipv4.dns "8.8.8.8 1.1.1.1" \ ipv4.method manual # Bring the
connection up sudo nmcli con up "Wired connection 1" # Verify ip addr show eth0
```

NOTE The gateway address (192.168.50.1) points to the Pi itself for the static assignment, not to the ASUS router. This is correct — the Pi is an application server, not a router. The ASUS router at the same IP handles all internet routing for the LAN.

If NetworkManager shows the connection name differently:

```
# List all connections nmcli con show # Use the exact name shown, e.g. "Wired connection 2"
nmcli con mod "Wired connection 2" ...
```

33.6 Verify the Full Stack

After completing hardware setup, run through this checklist to confirm everything is working before an activation.

CHECK	HOW TO VERIFY	EXPECTED RESULT
ASUS router up	Check Power LED on router	Solid white/blue LED
EMCOMM-NET visible	Scan Wi-Fi on any phone	EMCOMM-NET appears in the Wi-Fi list
Switch powered	Check Link LEDs on UniFi switch	Ports 1 and 2 show active link LEDs
Pi on correct IP	SSH to Pi: run <code>ip addr show eth0</code>	Shows 192.168.50.1/24
FieldComms reachable	Connect to EMMCOMM-NET, open browser to http://192.168.50.1	FieldComms dashboard loads
DHCP working	Check IP on a connected phone	Receives 192.168.50.100–200
Internet (if available)	Open http://192.168.50.1/propagation.html	Solar data loads (if WAN is connected)
APRS working	Open Tactical Map, check Sources panel	Graywolf shows Connected or station count > 0

33.7 Power Budget & Field Deployment Notes

DEVICE	TYPICAL DRAW	NOTES
Raspberry Pi 5 (16 GB) · Pironman MAX 5	10–15W	Pironman MAX 5 uses its own PSU. RAID 1 NVMe adds ~2–4W vs single drive.
ASUS RT-BE58 Go	8–10W	USB-C 18W PD; runs from a 20,000 mAh PD power bank for ~8 hours
UniFi Switch Flex 2.5G-5	3–6W	USB-C 5V/1A; can run from ASUS router's USB-A port or a phone charger
Total (typical)	16–28W	A 100Wh USB-C power bank (airline carry-on limit) gives 3–6 hours

TIP For extended operations, power the ASUS router and UniFi switch from separate USB ports on a power distribution strip rather than daisy-chaining from one bank. This ensures the network stays up even if the Pi's power bank is swapped mid-operation.

33.8 Extending Coverage — ASUS AiMesh

A single ASUS RT-BE58 Go covers approximately 2,000–2,500 sq ft under typical indoor conditions. For larger deployments — multi-room EOC buildings, large shelters, outdoor staging areas, or multi-floor facilities — Wi-Fi coverage can be extended using **ASUS AiMesh**. AiMesh adds one or more additional ASUS routers as wireless or wired nodes on EMMCOMM-NET. All nodes share the same SSID (EMMCOMM-NET), the same password, and the same 192.168.50.x subnet. Devices roam seamlessly between nodes with no reconfiguration required — the Pi remains at 192.168.50.1 regardless of which node a device is connected through.

Compatible AiMesh Node Hardware

ROUTER	FORM FACTOR	BEST USE
ASUS RT-BE58 Go (same as primary)	Travel / portable	Identical hardware — simplest pairing, battery-friendly, good for field extensions

ROUTER	FORM FACTOR	BEST USE
ASUS ZenWiFi Pro ET12	Tri-band desktop	Large indoor venues, EOC buildings with many simultaneous devices
ASUS RT-AX88U Pro	Desktop	High-density environments — high client count and throughput
Any AiMesh-capable ASUS router	Varies	Most ASUS routers released since 2018 support AiMesh — check asus.com/aimesh

Adding an AiMesh Node — Step by Step

- 1. Factory reset the node router** if previously configured. Hold the reset button for 5–10 seconds until the power LED flashes.
- 2. Power on the node** and place it within 30 ft of the primary router for initial pairing — it can be moved to its final position after.
- 3. On the primary router** (<http://192.168.50.1>), go to: **AiMesh** → **Add AiMesh Node**. The router scans for nearby compatible nodes.
- 4. Select the new node** from the detected list and click **Connect**. The primary router automatically pushes the EMCOMM-NET SSID, password, and 192.168.50.x subnet configuration to the node. This takes 1–3 minutes.
- 5. Move the node** to its final position once pairing completes. The node connects back to the primary via wireless backhaul automatically.
- 6. Test coverage** by walking the area with a phone connected to EMCOMM-NET and confirming <http://192.168.50.1> loads throughout the deployment space.

Wireless vs Wired Backhaul

BACKHAUL TYPE	SETUP	WHEN TO USE
Wireless backhaul (no extra cable)	Default after pairing — no additional setup needed. Node connects to primary via Wi-Fi automatically.	When running cable is impractical — outdoor areas, across hallways, temporary or rapid deployments
Wired Ethernet backhaul (recommended)	Run a CAT 6 cable from any UniFi switch port to the node's LAN port. AiMesh detects the wired connection automatically within 60 seconds.	Permanent or semi-permanent EOC setups. Fastest and most reliable — frees up Wi-Fi spectrum for client devices

Deployment Scenarios

SCENARIO	RECOMMENDED SETUP
Single room EOC ($\leq 2,500$ sq ft)	1× RT-BE58 Go primary only — no extension needed
Multi-room EOC or large shelter (2,500–7,500 sq ft)	1× RT-BE58 Go primary + 1 AiMesh node (wireless or wired backhaul)
Large building or campus (7,500–20,000 sq ft)	1× primary + 2–3 nodes, wired backhaul recommended for all nodes
Outdoor SAR staging area	1× primary at command post + battery-powered RT-BE58 Go nodes at field positions and medical staging area

TIP Up to 8 AiMesh nodes are supported on most ASUS routers. You do not need to pre-purchase nodes for every deployment — add them only when coverage is insufficient. Pairing a new node takes under 5 minutes on site.

35 JS8Call — HF Digital Keyboard Messaging (Windows)

JS8Call is a weak-signal HF digital mode designed for keyboard-to-keyboard messaging, store-and-forward relay, and group nets — all without an internet connection. It runs on the **Windows laptop** alongside Winlink Express, connected to the IC-7300 via USB. The FieldComms dashboard provides a quick-launch card that opens JS8Call's built-in web interface from any device on EMCOMM-NET.

35.1 What JS8Call Does

CAPABILITY	DESCRIPTION
Keyboard-to-keyboard	Type messages directly to any JS8 station within range — no internet required
Store and forward	Messages are relayed station-to-station across multiple hops when no direct path exists
Group / net messaging	Send a message addressed to a group callsign (e.g. @EMCOMM); any station in the group receives it
Heartbeat / beacon	Periodically transmits your callsign and grid square so nearby stations know you are on
APRS gateway	JS8Call can pass APRS messages via HF to the internet APRS-IS network when available
API interface	Exposes a TCP API on port 2442 that the FieldComms dashboard card connects to

TIP JS8Call complements Winlink — Winlink is best for structured formal messages and ICS forms; JS8Call is best for informal situational awareness traffic, check-ins, and keyboard conversation when Winlink is not the right tool.

35.2 Installation on the Windows Laptop

- 1. Download JS8Call** from <http://js8call.com>. Select the Windows installer (js8call-X.X.X-win64.exe). Run the installer and accept the defaults.
- 2. Audio setup:** Connect the IC-7300 via USB. In JS8Call, go to **File** → **Settings** → **Audio**. Set **Input** to the IC-7300 USB audio device (same device used for Winlink/VARA — you will switch between them). Set **Output** to the same IC-7300 USB audio device.
- 3. Radio / CAT setup:** Go to **File** → **Settings** → **Radio**. Set **Rig** to **Hamlib NET rigctl** if using OmniRig or WSJT-X CAT sharing, or set **Rig** to **Icom IC-7300** directly. Set **PTT Method** to **CAT**. Set the COM port to match the IC-7300's CI-V port (same port as Winlink — only one application can use it at a time). Set **Baud Rate** to **115200**.
- 4. Callsign and grid:** Go to **File** → **Settings** → **General**. Enter your **Callsign** (e.g. K9ESV) and your **Maidenhead grid square** (e.g. EN52 for McHenry County). This is what JS8Call broadcasts in its heartbeat.
- 5. Operating frequency:** Set the VFO on the IC-7300 or in JS8Call to the JS8 calling frequency for your primary band. Common frequencies: **7.078 MHz (40m)** — most active, primary EmComm choice; **14.078 MHz (20m)** — regional/national coverage; **3.578 MHz (80m)** — local/regional at night.
- 6. Mode speed:** Go to **Mode** menu and select **Normal** (default, ~16 WPM). Use **Slow** for very weak signals; **Fast** or **Turbo** when conditions are good and speed matters.

35.3 Enabling the TCP API (Required for FieldComms Integration)

JS8Call exposes a TCP API on port 2442 that allows the FieldComms dashboard card to open the JS8Call interface from any device on EMCOMM-NET. You must change the hostname from localhost to allow network access.

1. In JS8Call, go to **File** → **Settings** (or press **F2**).
2. Click the **Reporting** tab.
3. Under the **API** section, make the following changes:
4. Set **TCP Server Hostname** to **0.0.0.0** (this allows any device on the network to connect — the default 127.0.0.1 only allows the local machine).
5. Check **Enable TCP Server API**.
6. Verify **TCP Server Port** is set to **2442**.
7. Check **Accept TCP Requests**.
8. Also check **Allow setting station information from network** (required for some integrations).
9. Click **OK**. Restart JS8Call for the API settings to take effect.

NOTE Changing the TCP hostname to 0.0.0.0 means any device on EMCOMM-NET can reach JS8Call's interface. This is intentional — it is what allows operators at the EOC to open JS8Call from their phones or tablets via the FieldComms dashboard card. On a closed EMCOMM-NET network with no internet this is safe.

35.4 Windows Firewall — Allow JS8Call Through

Windows Firewall will block incoming connections to port 2442 by default. You must add a rule to allow it.

1. Open **Windows Defender Firewall with Advanced Security** (search "Windows Firewall" in the Start menu, click Advanced Settings).
2. Click **Inbound Rules** → **New Rule**.
3. Select **Port**, click Next.
4. Select **TCP**, enter **2442** as the specific port, click Next.
5. Select **Allow the connection**, click Next.
6. Check **Private** and **Domain** (uncheck Public if desired for security), click Next.
7. Name the rule **JS8Call API**, click Finish.

TIP Alternatively, when JS8Call first tries to listen on port 2442 Windows may show a "Windows Security Alert" popup asking if you want to allow access — click **Allow access** and the firewall rule is created automatically.

35.5 Find the Windows Laptop IP Address

The FieldComms dashboard card needs the IP address of the Windows laptop on EMCOMM-NET. Run this command in a Command Prompt on the Windows machine:

```
ipconfig
```

Look for the **Ethernet adapter** or **Wi-Fi adapter** section showing the EMCOMM-NET connection. The **IPv4 Address** will be in the 192.168.50.x range — for example **192.168.50.105**. Note this address.

TIP To avoid having to look this up every deployment, assign the Windows laptop a static IP via the ASUS router's DHCP reservation feature. Go to ASUS Web UI → LAN → DHCP Server → Manually Assigned IP, enter the laptop's MAC address and assign it a fixed address such as 192.168.50.2. It will always get the same IP when connected to EMCOMM-NET.

35.6 Configure the FieldComms Dashboard Card

1. On any device connected to EMCOMM-NET, open the FieldComms dashboard at **http://192.168.50.1**.
2. In the **Amateur Radio** mode section, find the **JS8Call** card (purple).

3. Tap or click the card. A prompt appears asking for the Windows laptop's IP address.
4. Enter the IP address noted from Step 35.5 — for example **192.168.50.105**.
5. Click OK. The card updates to show the IP address and immediately opens JS8Call's web interface at **http://192.168.50.105:2442** in a new tab.
6. The IP is saved on that device. Future taps open JS8Call directly without prompting.
7. To change the IP later (e.g. the laptop got a different address), tap the card again and enter the new IP, or leave the field blank to clear it.

NOTE Each device on EMCOMM-NET stores the JS8Call IP in its own browser localStorage. If a new operator connects with a different device, they will need to tap the card once to enter the IP. This is a one-time step per device per deployment.

35.7 JS8Call Operating Basics

ELEMENT	DESCRIPTION
Band Activity window	Shows all decoded JS8 signals heard — scroll up to see older entries
Call Activity window	Shows stations that have heard you or called you recently
Waterfall	Visual display of the HF band — JS8 signals appear as diagonal streaks
Offset	Your TX frequency within the audio passband (500–2500 Hz) — click the waterfall to tune
TX window	Type your message here — click TX or press Ctrl+Enter to transmit
Heartbeat	Automatic periodic transmission of your callsign/grid — enable under Mode menu
Speed indicator	Normal / Slow / Fast / Turbo — shown in lower right; Normal is standard for EmComm
SNR	Signal-to-noise ratio in dB shown next to decoded calls — JS8 works reliably at -20 dB SNR

35.8 EmComm Operating Conventions

1. **Monitor first:** Before transmitting, watch the Band Activity window for 2–3 minutes to learn what is on the frequency and who else is operating.
2. **Use directed messages:** Address messages to a specific callsign (e.g. W9ABC: CHECKING IN, EOC OPERATIONAL) rather than CQ to keep traffic efficient during activations.
3. **Group addressing:** Use a group callsign agreed upon before deployment — e.g. @EMCOMM or @K9ESV — so all stations in the group receive the message simultaneously.
4. **Heartbeat:** Enable the heartbeat so your station appears in other stations' call activity windows passively. Use **Normal** speed; disable during heavy traffic periods to save airtime.
5. **Coordinate with Winlink:** JS8Call and Winlink Express cannot use the IC-7300 simultaneously. Establish a schedule — for example, Winlink traffic at the top of the hour, JS8Call monitoring in between — or use a second radio for one of the modes.
6. **Logging:** JS8Call logs all activity to a local file. For incident documentation, export the log (File → Open Log Directory) and archive it with the incident record.

35.9 Sharing the IC-7300 with Winlink / VARA

Both JS8Call and Winlink Express need the IC-7300's USB audio and CI-V CAT port. Only one can be active at a time. Manage this with a simple protocol:

SCENARIO	PROCEDURE
Switching from Winlink to JS8Call	1. Close VARA HF (or wait for transmission to finish). 2. Close or minimize Winlink Express. 3. In JS8Call, go to File → Settings → Radio and confirm the COM port is correct. 4. Click Connect in JS8Call. Verify CAT control is working (frequency shows in JS8Call).
Switching from JS8Call to Winlink	1. Click Disconnect in JS8Call (or close JS8Call). 2. Open VARA HF. It will reclaim the audio device. 3. Open Winlink Express and connect as normal.
Running both simultaneously (two radios)	Connect a second HF radio to the laptop via a separate USB audio and CAT interface. Configure JS8Call to use the second radio's audio/CAT port. Both can run at the same time with no conflicts.

TIP OmniRig (free, from Afreet Software) can share a single CAT port between multiple applications simultaneously. Install OmniRig, configure it for the IC-7300, then set both Winlink Express and JS8Call to use OmniRig as the CAT interface. Audio switching is still manual.

35.10 Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
JS8Call card opens blank page or "Connection refused"	JS8Call not running, API not enabled, or firewall blocking port 2442	Start JS8Call on Windows. Verify API is enabled (Section 35.3). Check Windows Firewall (Section 35.4).
Card shows wrong IP or does not connect	Windows laptop got a different IP from DHCP	Run ipconfig on laptop, tap the JS8Call card again and update the IP. Set a DHCP reservation in the ASUS router to fix permanently (Section 35.5).
No audio decoded — waterfall is flat	Wrong audio input selected, IC-7300 USB audio not recognized, or radio in wrong mode	In IC-7300, confirm mode is USB-D. In JS8Call Settings → Audio, confirm IC-7300 USB Codec is selected as Input.
CAT control error on startup	COM port conflict with Winlink/VARA or wrong baud rate	Close VARA HF and Winlink Express before starting JS8Call. Confirm baud rate 115200 in Settings → Radio.
Transmitting but no one hears	Low TX level, wrong frequency, or antenna issue	Check IC-7300 ALC — should show movement but not peaking. Verify operating on 7.078 MHz. Check antenna SWR.
API connects but no messages showing	Normal — no JS8 activity on the band right now	Wait and monitor. Try 40m (7.078 MHz) if on another band. Enable Heartbeat to prompt responses.

36 ICS Planning P — Operational Planning Cycle

<http://192.168.50.1/ics/planningp.html>

The Planning P page is an interactive guide to the ICS operational planning cycle. It presents all 15 phases of the Planning P — from the initial incident through each operational period — with the standard agenda, required ICS forms, and attendee list for every phase. Access it from the **■ Planning P** tab on any ICS platform page.

The Planning P at a Glance

The Planning P is the standard ICS planning cycle diagram that shows how each operational period is developed. The vertical stem represents the initial response phases. The loop represents the repeating planning cycle for each subsequent operational period.

The 15 Phases

PH ASE	NAME	KEY FORMS
1	Incident / Event	ICS-201 (begin)
2	Notification	ICS-201
3	Initial Response & Assessment	ICS-201
4	Initial UC Meeting	ICS-201, ICS-202
5	Incident Brief — ICS-201	ICS-201
6	IC/UC Develop/Update Objectives Meeting	ICS-202
7	Command & General Staff Meeting/Briefing	ICS-202, ICS-203
8	Preparing for the Tactics Meeting	ICS-215, ICS-215A
9	Tactics Meeting	ICS-215, ICS-215A, ICS-203, ICS-204
10	Preparing for the Planning Meeting	ICS-204, ICS-205, ICS-206, ICS-207, ICS-208
11	Planning Meeting	ICS-202 through ICS-206
12	IAP Prep & Approval	ICS-202 through ICS-208 (complete IAP)
13	Operations Briefing	Complete IAP, ICS-204 by Division
14	Execute Plan & Assess Progress	ICS-214, ICS-209, ICS-309
15	New Ops Period Begins	ICS-214, ICS-209

Using the Planning P Page

1. Open the ICS Platform (<http://192.168.50.1/ics/>) and click the **■ Planning P** tab.
2. The reference image of the official Planning P diagram is shown on the left for reference.

3. The 15 phase buttons are listed in the center panel, organized into five color-coded groups: gray (Initial Response), yellow (Establish Objectives), red (Develop the Plan), green (Prepare & Disseminate), and teal (Execute & Evaluate).
4. Click any phase button to see its details in the right panel.

Phase Detail Panel

Clicking a phase button opens its detail panel on the right showing three sections:

SECTION	WHAT IT SHOWS
Standard Agenda	Numbered list of the standard agenda items for that meeting or activity
Required Forms	ICS form numbers as clickable chips — click to open the form directly
Who Should Attend	List of ICS roles required or optional for that phase, with red/green roster dots showing who is currently checked in

Generating a Briefing Sheet

1. Select any phase by clicking its button.
2. Click **Generate briefing sheet** in the detail panel.
3. A printable one-page cover sheet opens in a new window with: the phase name, agenda, required forms, and an attendance table with signature lines.
4. Print it for the IAP package or the planning meeting folder.

The Phase Groups and Their Colors

COLOR	GROUP	PHASES	ICS PHASE OF PLANNING
Gray	Initial Response	1–5	Stem — initial incident response
Yellow	Establish Objectives	6–7	Stem — incident objectives set by IC/UC
Red	Develop the Plan	8–11	Loop — tactics through planning meeting
Green	Prepare & Disseminate	12–13	Loop — IAP approval and ops briefing
Teal	Execute, Evaluate & Revise	14–15	Loop — field operations and next period

Who Attends Each Phase

The attendee list for each phase is drawn from standard ICS doctrine. The red/green status dots cross-reference the active incident roster — green means that role is filled by a checked-in member, red means the role is unfilled. This gives the IC an immediate gap identification tool before each meeting.

PHASE	REQUIRED ATTENDEES (KEY ROLES)
IC Briefing (Phase 5)	IC, all Section Chiefs, Safety Officer
C&G; Staff Mtg (Phase 7)	IC, all Section Chiefs, Command Staff
Tactics Meeting (Phase 9)	IC, OSC, PSC, Safety Officer, Resource Unit Leader, Situation Unit Leader
Planning Meeting (Phase 11)	IC, all Section Chiefs, Safety Officer, Resource Unit Leader

PHASE	REQUIRED ATTENDEES (KEY ROLES)
IAP Prep (Phase 12)	IC, PSC, Safety Officer, Documentation Unit Leader
Ops Briefing (Phase 13)	All incoming resources, IC, all supervisors

TIP The Planning P page is a guide, not a workflow enforcer. You do not need to click through the phases in order. Use it as a quick reference during planning meetings to ensure nothing is missed — especially on activations where the planning cycle is compressed.

34 Appendix — Administration & Quick Reference

A1 Installation & Updates

Incident Management is installed and updated with scripts on the Pi. These are net-manager tasks, run over SSH or with a keyboard attached to the Pi.

COMMAND	WHAT IT DOES
<code>sudo bash install.sh</code>	Interactive installer — sets callsign, coordinates, Wi-Fi, services
<code>sudo bash update.sh</code>	Updates Incident Management code and restarts services
<code>sudo bash kiwix_setup.sh</code>	Install/manage Kiwix offline content (Chapter 31)
<code>sudo bash download_tiles.sh</code>	Download offline map tiles (Chapter 32)

The installer patches your station callsign (K9ESV) and coordinates into the pages, sets the static AP address 192.168.50.1, and enables all services to start on boot. The install log is at `/var/log/fieldcomms-install.log`.

A2 Wi-Fi Access Point Configuration

Wi-Fi is provided by the **ASUS RT-BE58 Go** travel router — the Pi no longer runs hostapd. Configure the router as described in Chapter 34. Quick reference:

SETTING	VALUE	WHERE TO CHANGE
SSID	EMCOMM-NET	ASUS Web UI → Wireless → General
Password	Set during router setup	ASUS Web UI → Wireless → General
Router LAN IP	192.168.50.1	ASUS Web UI → LAN → LAN IP
DHCP range	192.168.50.100–200	ASUS Web UI → LAN → DHCP Server
Pi static IP	192.168.50.1	nmcli on Pi (see Chapter 34.5)
Aliases	fieldcomms.local, emcomm.local	dnsmasq on Pi (if enabled)

IMPORTANT If the ASUS router is factory-reset in the field, its LAN will revert to 192.168.1.x and FieldComms will be unreachable until you reconfigure the subnet to 192.168.50.x. See Chapter 34.1 for the full setup procedure.

A3 Data Storage & Backup

DATA	LOCATION
Main database (nets, roster, resources...)	<code>/opt/fieldcomms/data/fieldcomms.db</code>
FCC license database (~600 MB)	<code>/opt/fieldcomms/data/fcc.db</code>
Reference Library files	<code>/opt/fieldcomms/data/refs/</code>
Map tiles	<code>/opt/fieldcomms/tiles/</code>
Kiwix ZIM files	<code>/opt/kiwix/zim/</code>

The main database is a single SQLite file. Back it up by copying fieldcomms.db to a USB drive. Export the roster CSV monthly (Chapter 4) and keep a copy on a USB drive labeled FIELDCOMMS as a secondary backup.

A4 Winlink — Email Over Radio

MCESV/MCEMA uses **Winlink Express** as the primary Winlink client. It runs on a dedicated **Windows station** (laptop or PC) connected to the radio, and it is the client our operators are trained on and use day to day. Express has the fullest mode support, including VARA HF and VARA FM, and is the recommended choice for any operator running Winlink traffic during an activation.

Pat — Browser-Based Backup on the Pi

The Incident Management server also runs **Pat**, an open-source Winlink client, on the Pi at port 8090. Pat is the **backup** Winlink path. Its advantage is that it is reached through a web browser, so any device on EMCOMM-NET — including a phone or tablet with no Winlink software installed — can send and receive Winlink mail through it. Use Pat when the Windows Express station is unavailable, or when the only device on hand is a tablet.

Which to Use

SITUATION	USE
Normal operations, Windows station available	Winlink Express on the Windows station (primary)
Windows station down or unavailable	Pat on the Pi via browser (port 8090)
Only a phone or tablet on EMCOMM-NET	Pat on the Pi via browser
Need full VARA HF / VARA FM performance	Winlink Express on the Windows station

NOTE The Windows Express station is a separate machine that the Incident Management server does not manage or monitor — set it up and test it independently as part of your pre-flight check. The Pi only runs and monitors Pat (the backup). Both can share one radio via a switch, or use separate radios.

A5 Setting Up the Windows Express Station

This section sets up the primary Winlink station — a Windows PC or laptop running Winlink Express with the VARA modem. Do this once per station, well before an activation, and test it. Each operator running this station should have done it on their own machine or be trained on the shared station.

What You Need

- A Windows PC or laptop (Express is Windows-only).
- A radio interface that connects the PC sound and PTT to the radio (for example a SignalLink, a Digirig, or a radio with a built-in USB sound device).
- The radio, with the correct CAT/PTT cabling if using rig control.
- An amateur radio license and callsign (each operator uses their own).

Step 1 — Install Winlink Express

1. On the Windows PC, go to winlink.org/WinlinkExpress and download the installer.
2. Run the installer and follow the prompts. The Express icon is a purple-gray checkerboard.
3. Launch Express. On first run it opens **Settings** → **Winlink Express Setup**.
4. Enter your **callsign**, your **6-character grid square** (e.g. EN52), your contact information, and a **password-recovery email** (use a non-Winlink address). Click Update.

NOTE Your Winlink account is created automatically the first time you successfully send a message. Accounts that go unused for an extended period are removed from the servers, so send at least one message periodically to keep the account active.

Step 2 — Test Over the Internet First (Telnet)

Before involving the radio, prove the software and account work over the internet.

1. In the **Open Session** dropdown (upper right), choose **Telnet Winlink**.
2. Click **Message** → **New Message**, address it to your own callsign, type a short test, and click **Post to Outbox**.
3. Click **Open Session**, then **Start**. The message uploads and your account is created.
4. This confirms Express and your account are working before you add radio complexity.

Step 3 — Install the VARA Modem

VARA is the modem software Express uses over the radio. VARA HF is for HF; VARA FM is for VHF/UHF. Install whichever modes your station will use.

1. Download VARA HF and/or VARA FM from the author's site: rosmodem.wordpress.com.
2. Run the installer(s). VARA HF installs to C:\VARA by default; VARA FM has its own installer.
3. If your antivirus flags VARA as a threat, this is a known **false positive** — create an exception for it.
4. The unregistered (free) version works but is speed-limited. A one-time registration key from the author unlocks full speed; the team may register the shared station.

Step 4 — Connect Express to VARA and the Radio

1. In Express, set the **Open Session** dropdown to **VARA HF Winlink** (or VARA FM Winlink).
2. Click **Open Session** to launch the VARA TNC window. The Virtual TNC command port defaults to **8300** — it must match in both Express and the VARA app.
3. In VARA, open **Settings** → **Soundcard** and select your radio interface for input and output (e.g. the Signalink or USB Audio CODEC device).
4. In VARA, open **Settings** → **PTT** and set how the radio is keyed — VOX for a Signalink, or a COM/CAT port for rig control.
5. In Express, open **Settings** → **Radio Setup** and configure CAT/PTT if you are using rig control rather than VOX.

TIP Rig control (keying the radio) is the job of Winlink Express, not the VARA modem — VARA only generates and decodes the audio. If the radio will not transmit, check the PTT/CAT settings in Express first.

Step 5 — Make a Real Radio Contact

1. In Express, open a VARA session and use the channel selector / RMS list to find a nearby Winlink gateway (RMS) station. 80m and 40m are the most reliable HF bands over time.
2. Set the radio to the gateway's frequency, click **Start**, and let Express connect and exchange any queued messages.
3. Send a test message to your own callsign over the radio to confirm the full path works.

Sharing One Radio With Pat (the Pi Backup)

If the Windows Express station and the Pi (running Pat) share a single radio, only one may transmit at a time. Options:

- Use a manual A/B switch or operating procedure so only one client is connected to the radio at a time — this is the simplest and safest approach.
- Or give each client its own radio, eliminating any conflict.

- Never leave both clients in an active session on the same radio — they will key over each other.

NOTE Standard practice for many teams: Windows Express + VARA on HF as the primary station, and Pat on the Pi for VHF Packet or as the browser-accessible backup. Decide your radio assignment before an activation and write it into your station's operating procedure.

IC-7300 Specific Setup

The IC-7300 is an excellent Winlink radio because its internal sound card, CAT control, and PTT keying all travel over a **single USB cable** — no interface box, no SignalLink, no extra cables. Follow this sequence for a clean setup.

What You Need

- IC-7300 with power supply and antenna connected.
- A **shielded, high-quality USB-A to USB-B cable**. A cheap or unshielded cable causes interference that shows up as "birdies" in the receive waterfall. Use a name-brand cable.
- The **Icom IC-7300 USB driver** (Silicon Labs CP210x) installed on the Windows PC. Download from icom.america.com if not already installed.
- Winlink Express and VARA HF installed (see the steps above in this section).

Step 1 — Configure the IC-7300 Menu

On the radio: press **MENU** → **SET** → **Connectors**. Set these values:

MENU ITEM	VALUE
ACC/USB Output Select	AF
ACC/USB Output Level	50%
ACC/USB AF SQL	OFF (Open)
ACC/USB AF Beep/Speech Output	OFF
USB MOD Level	40%
USB Serial Function	CI-V
DATA OFF MOD	MIC, ACC
DATA MOD	USB

Still in **Connectors**, tap **CI-V** and set:

CI-V SETTING	VALUE
CI-V Baud Rate	Auto
CI-V Address	94h ← leave at default, Express expects this
CI-V Transceive	ON
CI-V Output (for ANT)	OFF
CI-V USB Port	Unlink from [REMOTE]
CI-V USB Baud Rate	115200
CI-V USB Echo Back	ON
Inhibit Timer at USB Connection	ON

Return to **SET** → **Function** and set:

FUNCTION SETTING	VALUE
RF/SQL Control	RF+SQL
Tuner → PTT Start	ON ← auto-tunes when software keys the radio

IMPORTANT Also check: COMP (compression) OFF, RF Power 100%, MIC Gain 30%. High compression distorts digital modes.

Step 2 — Set Windows Sound Levels

The IC-7300 appears in Windows as **USB Audio CODEC**. Set its levels in Windows Control Panel → Sound:

DEVICE	WINDOWS LEVEL
Recording — Microphone (USB Audio CODEC)	~28 (not too hot — avoid overdriving)
Playback — Speakers (USB Audio CODEC)	~80

IMPORTANT If you only see COM ports in the VARA soundcard dropdown and no "USB Audio CODEC", the IC-7300 USB driver did not install correctly. Reinstall the Silicon Labs CP210x driver from the Icom website and reboot.

Step 3 — Configure Winlink Express for the IC-7300

1. Open Winlink Express. In the session-type dropdown (top right) choose **VARA HF Winlink**.
2. Click **Open Session** to launch VARA. In the VARA window go to **Settings** → **Radio Setup**.
3. Click **Select Radio Model** and choose **Icom 7300**.
4. Set **Icom Address** to **94** and mode to **USB Digital**.
5. Under **Radio Control Port**: select the COM port you recorded from Device Manager (e.g. COM8), set Baud to **115200**, and **uncheck** Enable RTS and Enable DTR.
6. Under **PTT Port**: set Serial Port to **CI-V**, leave Baud at default.
7. Click **Update** to save.

Step 4 — Configure VARA Soundcard

1. In the VARA window go to **Settings** → **SoundCard**.
2. Set **Device Input** to **Microphone (USB Audio CODEC)**.
3. Set **Device Output** to **Speakers (USB Audio CODEC)**.
4. Click **OK**.
5. The VARA waterfall should immediately show receive noise from the radio. If the waterfall is blank, the wrong audio device is selected.

Step 5 — Set the Radio Mode

On the IC-7300 front panel, set the mode to **USB-D** (USB Digital) before starting a VARA session. VARA will not key up correctly in plain USB or AM.

TIP When Winlink changes frequency automatically (channel selection), the IC-7300 follows via CAT — you do not need to touch the radio. If the radio does not follow, check that CI-V Transceive is ON and the correct COM port is selected in Express.

Quick Reference — IC-7300 + VARA Settings

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	8 GB LPDDR4X	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	TBD (current model)	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	✓ YES — Pi 5 chip, 8 GB RAM, current generation	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit (\$120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit (\$120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

501(c)(3) Grant Funding Note: MCEMV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	Pi 500 + Pi Monitor	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed
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■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.

ITEM	SETTING
Radio mode	USB-D (USB Digital)
USB cable	Single shielded USB-A to USB-B, radio to PC
Windows audio device	USB Audio CODEC (input ~28, output ~80)
VARA radio model	Icom 7300
Icom CI-V address	94 (0x94h)
COM port baud rate	115200
RTS / DTR	Both unchecked/disabled
PTT method	CI-V (via the USB cable — no separate PTT cable needed)
COMP (compression)	OFF
RF Power	100%
MIC Gain	30%
VARA TNC port	8300 (default)

A6 Page Quick-Reference

TOOL	URL	WHEN TO USE
Dashboard	/	System status, alerts, tool access
Net Control	/netcontrol.html	ARES/RACES amateur nets
Starcom Logger	/starcom.html	Public-safety Starcom nets
Observer	/observer.html	Read-only net view for agencies
Roster	/roster.html	Member management, activations
Resources	/resources.html	Track personnel, vehicles, equipment
Tactical Map	/tactical.html	APRS situational awareness
Resource Map	/resmap.html	Manual Starcom unit placement
Callsign Lookup	/callsign.html	Offline FCC license lookup
Dead Man's Switch	/deadmans.html	Monitor nets for inactivity
Pre-Flight	/preflight.html	GO/CAUTION/NO-GO before activation
ICS Platform	/ics/	Declared ICS incidents
NTS Radiogram	/nts.html	Formal ARRL traffic
ICS-213	/ics213.html	Written general messages
ICS-214	/ics214.html	Unit activity logs
Propagation	/propagation.html	HF band conditions

TOOL	URL	WHEN TO USE
Repeaters	/repeaters.html	Local repeater directory
Facilities	/facilities.html	EOC, shelter, hospital directory
Grid Square	/grid.html	Maidenhead grid conversion
Cheat Sheets	/cheatsheets.html	Phonetics, Q-codes, band plan
Print Center	/printcenter.html	Print any form or cover sheet
Reference Library	/refs.html	Document storage & retrieval
Kiwix	:8081/	Offline medical/technical reference

A7 Field Operator Tips

- Keep the EMCOMM-NET password posted at the EOC for arriving operators.
- Bookmark <http://192.168.50.1/netcontrol.html> on your phone before an activation.
- The observer page works on any smartphone — share it with the served-agency duty officer.
- Run the Pre-Flight checklist before every activation, not just major ones.
- Export the ICS-309 from Net Control at the end of each operational period — it is the official communications record.
- WikiMed in Kiwix is your fastest offline medical reference — faster than paper manuals.
- Leave the Tactical Map open on a dedicated tablet at the EOC; it refreshes every 30 seconds.
- Enter members' certifications in the Roster before any activation.
- Export the Roster CSV monthly and keep a backup on the USB drive labeled FIELDCOMMS.

NOTE For questions or issues during an activation, the Health Monitor at <http://192.168.50.1:5051/health> shows complete system status. Contact your net manager or Emergency Coordinator for operational questions.